

ROG-Dia Benchmark Report

Date: 2026-03-01

1. Evaluated Models

- **diar sortformer 4spk v1** ([nvidia/diar_sortformer_4spk-v1](#)) - [HuggingFace](#)
- **diar streaming sortformer 4spk v2** ([nvidia/diar_streaming_sortformer_4spk-v2](#)) - [HuggingFace](#)
- **diar streaming sortformer 4spk v2.1** ([nvidia/diar_streaming_sortformer_4spk-v2.1](#)) - [HuggingFace](#)
- **pyannote 3 1** ([pyannote/speaker-diarization-3.1](#)) - [HuggingFace](#)
- **pyannote community 1** ([pyannote/speaker-diarization-community-1](#)) - [HuggingFace](#)
- **reverb diarization v2** ([Revai/reverb-diarization-v2](#)) - [HuggingFace](#)
- **speaker diarization precision 2** ([pyannote/speaker-diarization-precision-2](#)) - [HuggingFace](#)

2. Executive Summary

Model	Collar	DER	Purity	Cover	Miss	FA	Conf	RTF	VRAM (GB)	Completed
speaker diarization precision 2	0.25	18.68	87.29	89.32	16.53	1.25	0.56	0.03	0.0	12/12
diar streaming sortformer 4spk v2	0.25	21.18	86.97	89.09	18.54	1.37	0.77	< 0.01	1.6	12/12
pyannote community 1	0.25	23.81	81.28	86.74	15.14	1.73	7.64	0.06	1.6	12/12
pyannote 3 1	0.25	23.89	81.51	86.60	15.14	1.73	7.66	0.06	1.6	12/12
diar streaming sortformer 4spk v2.1	0.25	28.24	86.11	84.68	22.31	1.22	3.71	< 0.01	1.6	12/12
reverb diarization v2	0.25	28.25	73.84	91.79	8.13	2.20	18.33	0.10	4.3	12/12
diar sortformer 4spk v1	0.25	57.51	80.69	48.43	19.41	2.47	7.60	0.03	104.4	8/12

Terminology & Methodology

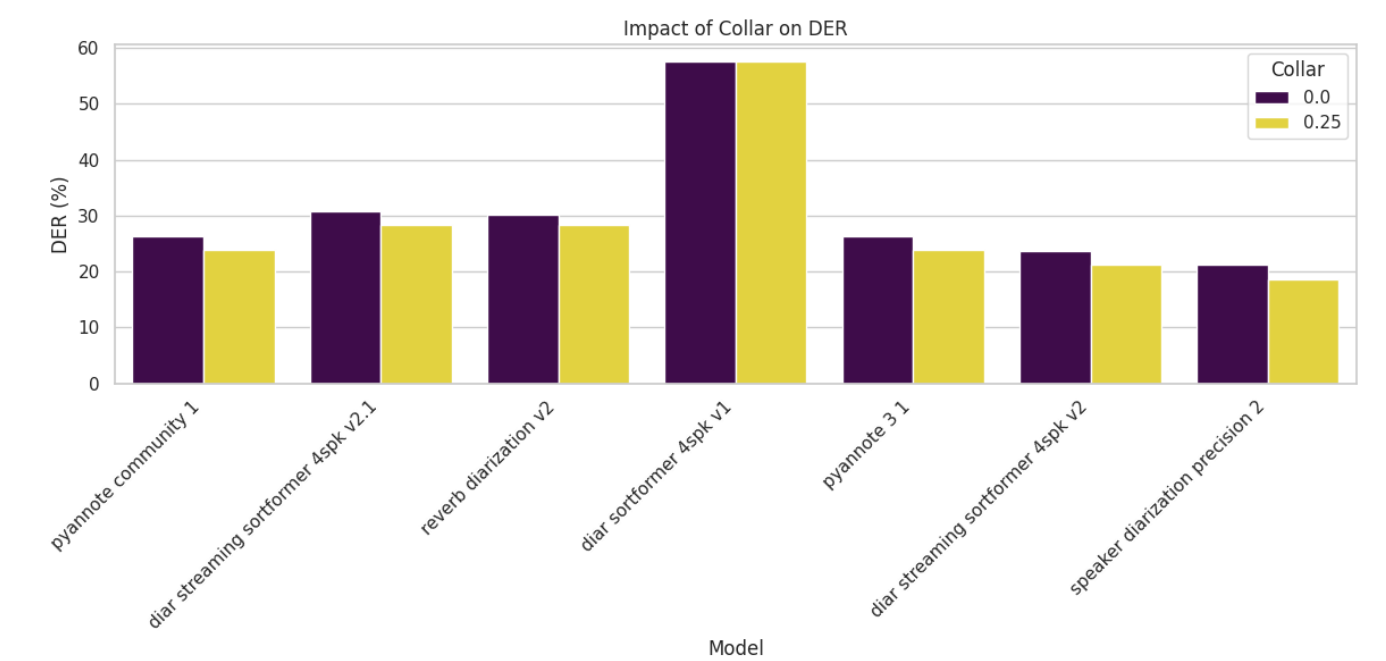
- **DER (Diarization Error Rate):** Primary metric. Lower is better. Sum of Missed, False Alarm, and Confusion rates.
- **Miss (%):** Speech present in Gold Standard but missed by the model.
- **FA (False Alarm %):** Model predicted speech where Gold Standard is silent.
- **Conf (Confusion %):** Speech correctly detected but assigned to the wrong speaker.
- **Purity (%):** Evaluates cluster purity. High purity = when a model identifies a speaker, it is consistently the same person.
- **Cover (Coverage %):** Evaluates how much of the original speaker's speech was captured under a single hypothesis cluster.
- **RTF (Real Time Factor):** Processing time divided by audio length. e.g., < 0.01 means exceptionally fast processing.
- **VRAM (GB):** Peak GPU memory utilized. 0.0 GB indicates an API/Cloud-based model.

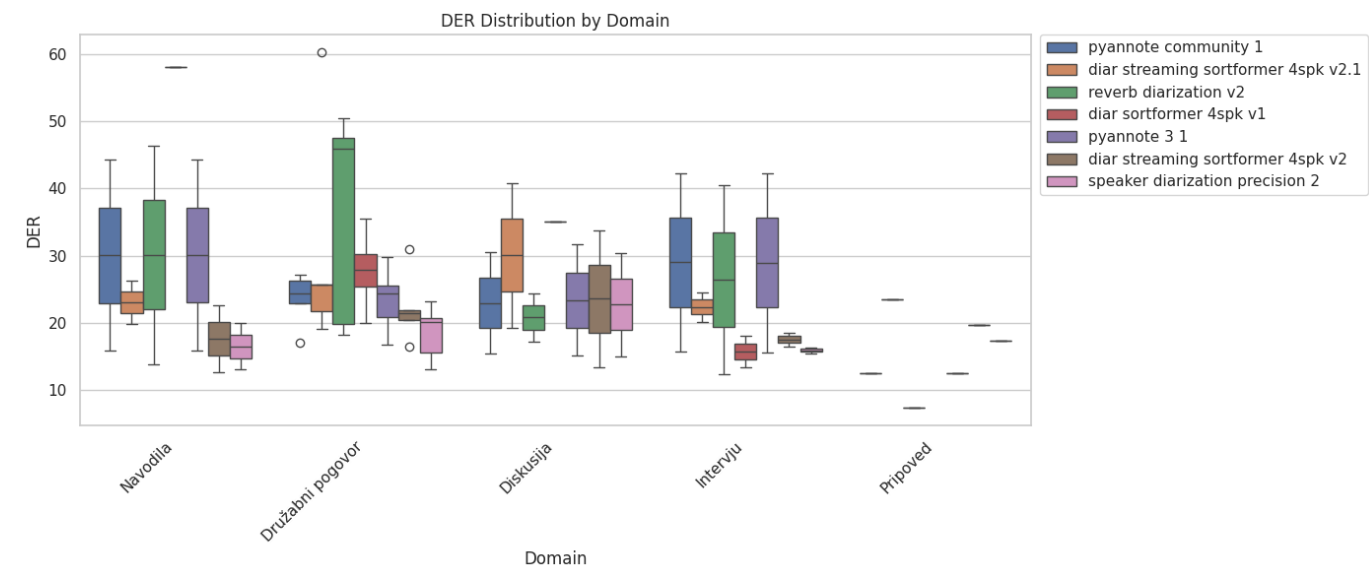
3. Dataset Errata (Corrections Applied)

Corrections automatically applied via Universal Evaluation Maps (UEM) to account for transcription errors. Models are not penalized for predictions outside these boundaries.

- **ROG-Dia-GS0-P0016:** Evaluated only up to 1172.092s. Reason: Zlati standard se zaključí predčasno pri 19:32 (potrdila avtorica). Preostanek posnetka se ignorira pri izračunu metrik.

4. Visual & Domain Analysis





Domain Comparison (DER %)

Average DER per domain. **Bold** highlights the best model for each domain.

Domain	A	B	C	D	E	F	G	AVG
Diskusija	35.09	23.59	30.05	23.36	22.95	20.78	22.74	25.51
Družabni pogovor	27.81	22.23	30.5	23.45	23.54	36.40	18.48	26.06
Intervju	15.70	17.47	22.34	28.96	29.02	26.42	15.89	22.26
Navodila	58.11	17.62	23.05	30.06	30.04	30.12	16.47	29.35
Pripoved	nan	19.63	23.43	12.42	12.45	7.31	17.37	15.44

- **A:** diar sortformer 4spk v1
- **B:** diar streaming sortformer 4spk v2
- **C:** diar streaming sortformer 4spk v2.1
- **D:** pyannote 3 1
- **E:** pyannote community 1
- **F:** reverb diarization v2
- **G:** speaker diarization precision 2

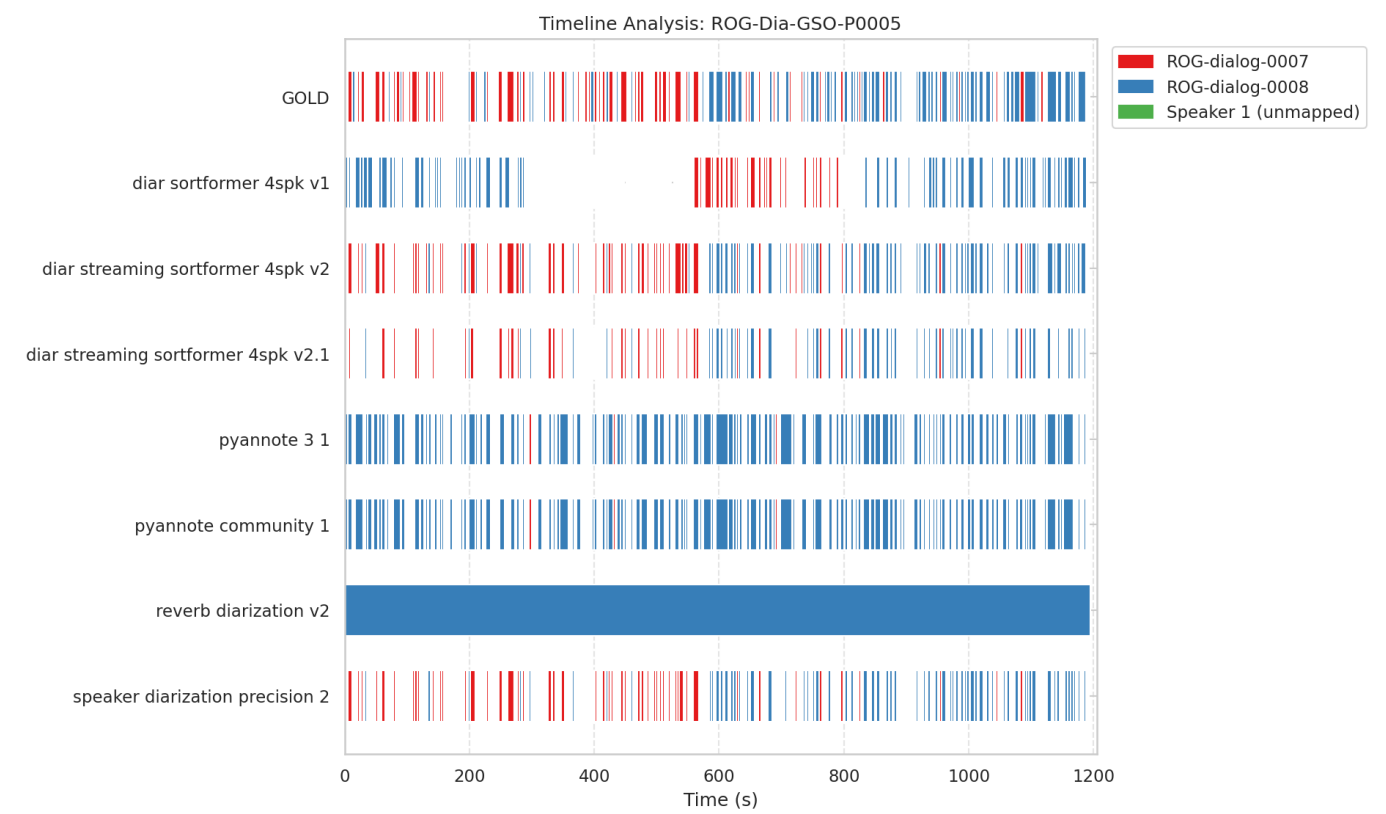
5. Deep Dive: File-by-File Analysis

Detailed breakdown for every file. *For metric definitions, see Executive Summary.*

File: ROG-Dia-GSO-P0005

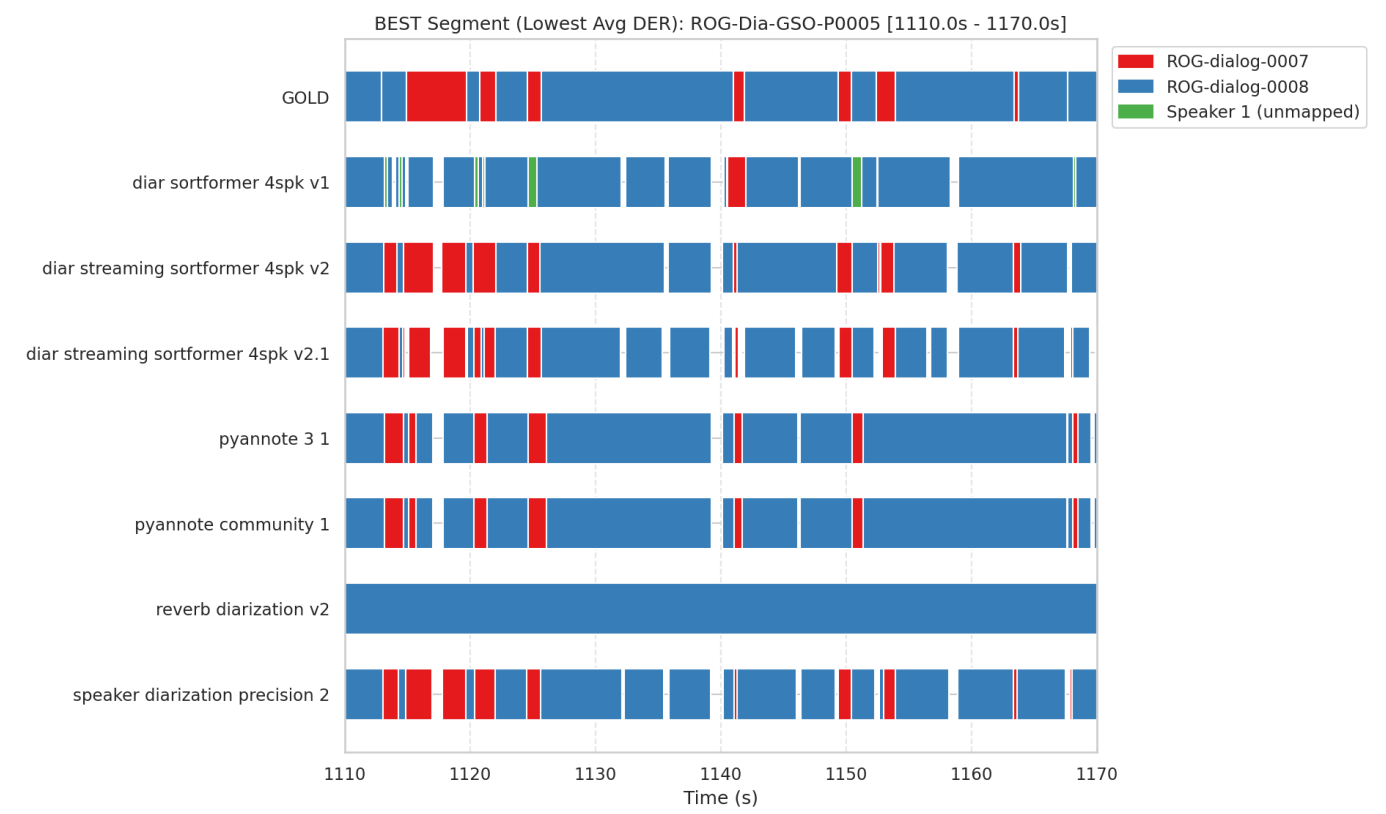
Domain: Navodila | **Quality:** 4 | **Device:** tablični računalnik Samsung Galaxy S6 Lite

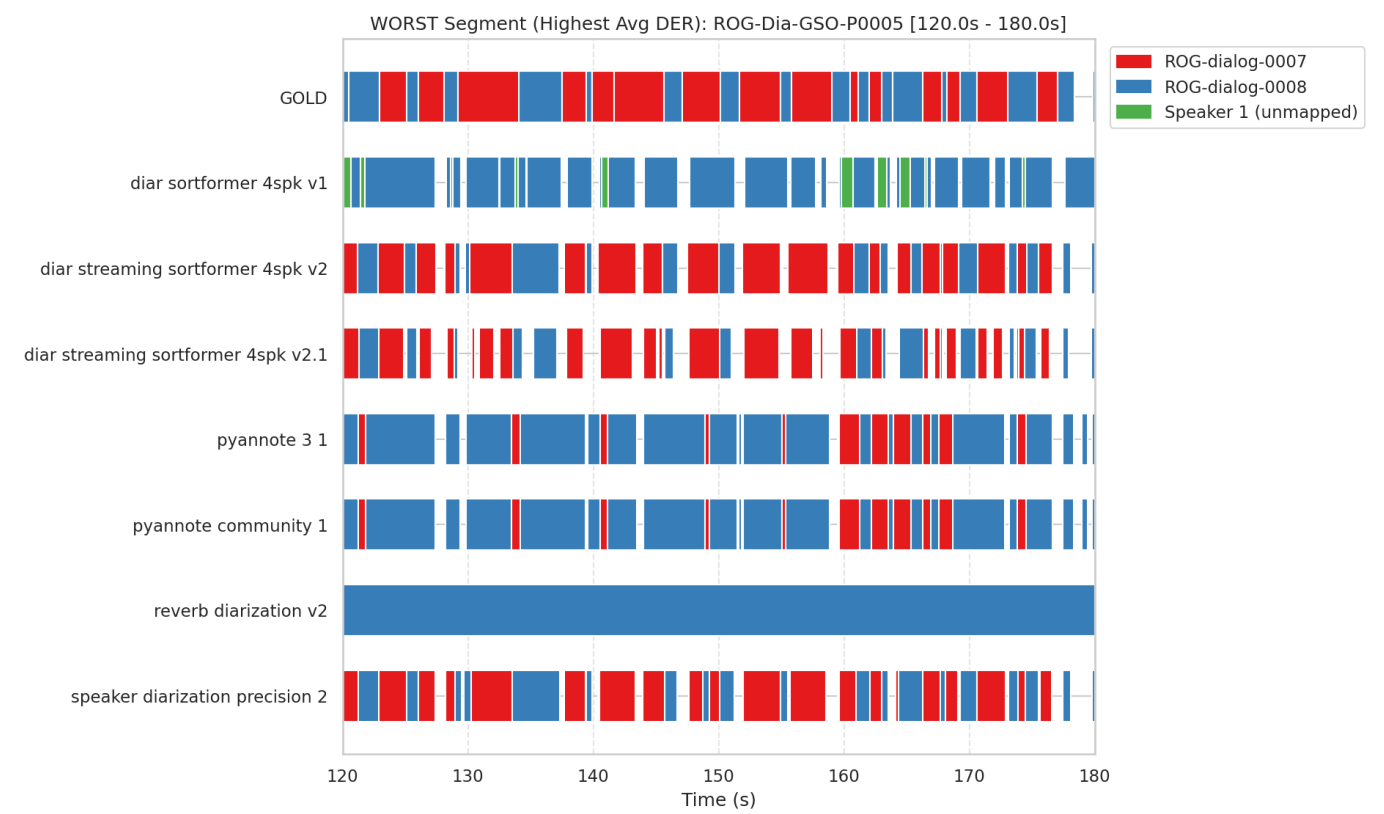
Z govorkama smo se predhodno dogovorile za snemanje. Da je bil pogovor sproščen, smo govor snemale na domu ene izmed njiju. Pogovor je potekal o temah, ki govorki zanimajo, da sta si lahko izmenjevali navodila.



60-Second Snippets (Zoom-in)

Below are 60-second zoomed-in windows showing where the models performed best and worst (based on average DER).



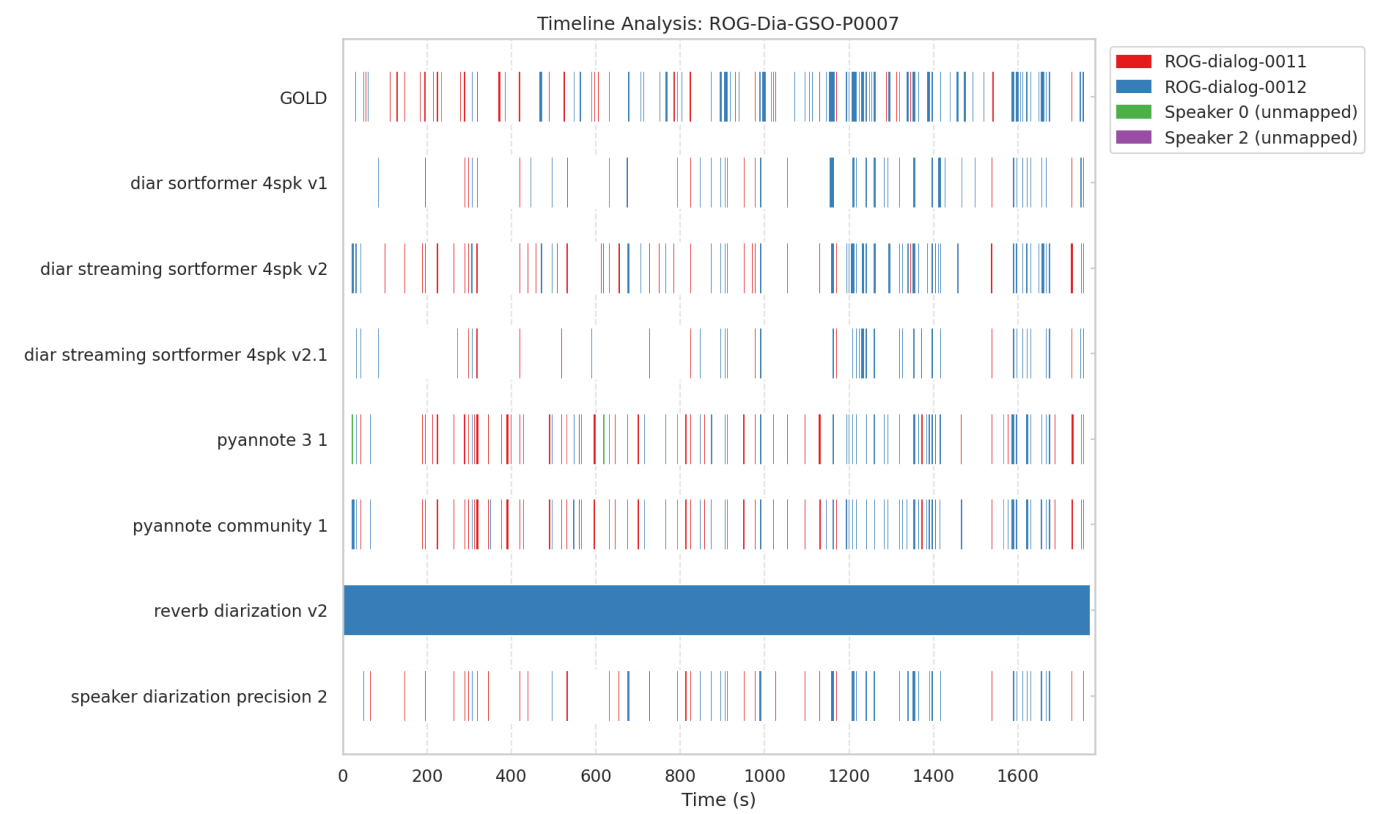


Model	DER	Miss	FA	Conf	Pur	Cov	VRAM (GB)
diar streaming sortformer 4spk v2	12.66	9.77	2.16	0.73	81.54	83.05	1.1
speaker diarization precision 2	13.00	10.95	1.65	0.40	82.12	83.87	0.0
diar streaming sortformer 4spk v2.1	19.80	18.07	1.15	0.58	82.60	85.81	1.1
pyannote community 1	44.24	9.05	3.73	31.45	53.49	85.79	1.6
pyannote 3 1	44.24	9.05	3.73	31.45	53.49	85.79	1.6
reverb diarization v2	46.39	11.11	0.56	34.72	54.17	100.00	4.3
diar sortformer 4spk v1	58.11	20.93	2.83	34.36	57.24	62.31	47.9

File: ROG-Dia-GSO-P0007

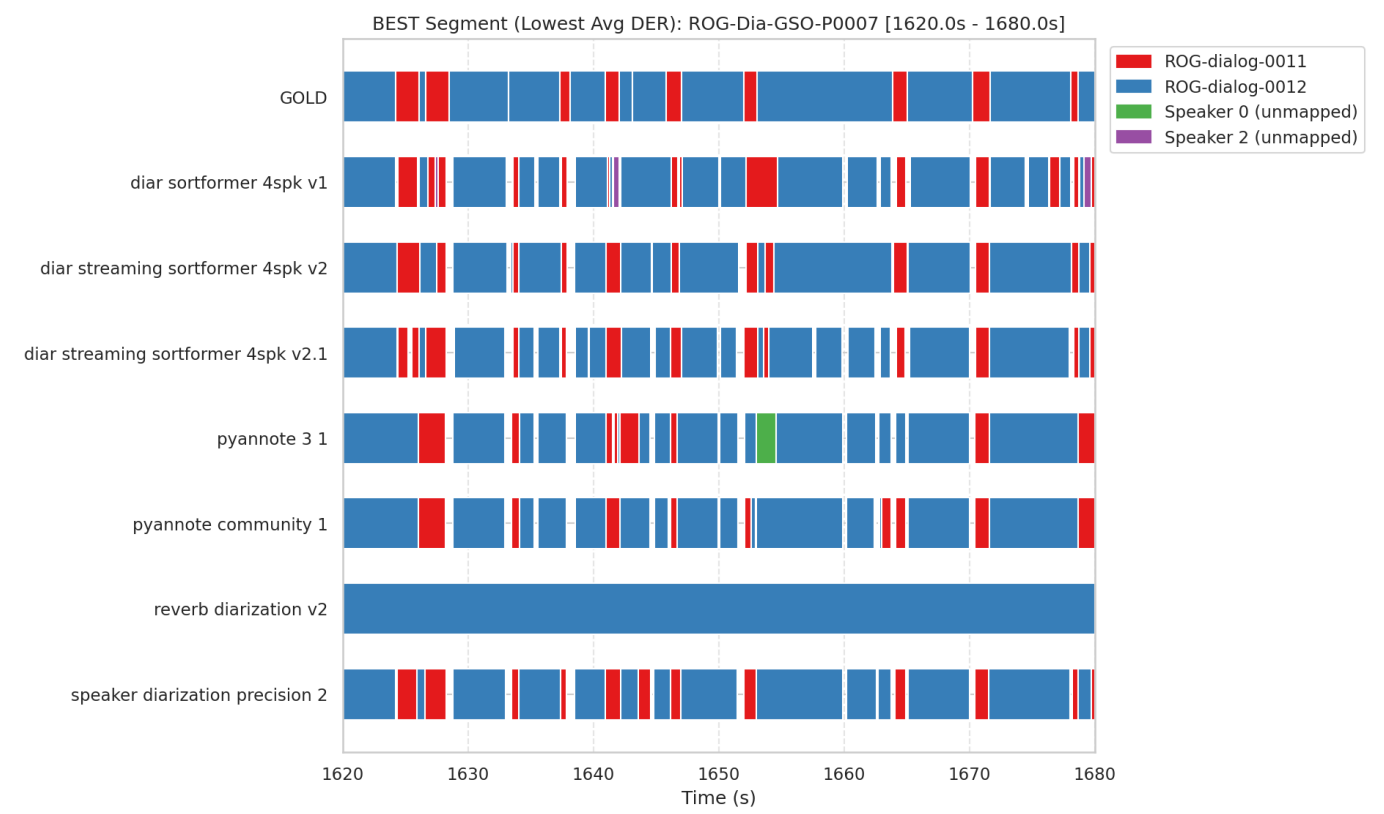
Domain: Družabni pogovor | Quality: 5 | Device: Apple iPhone 13

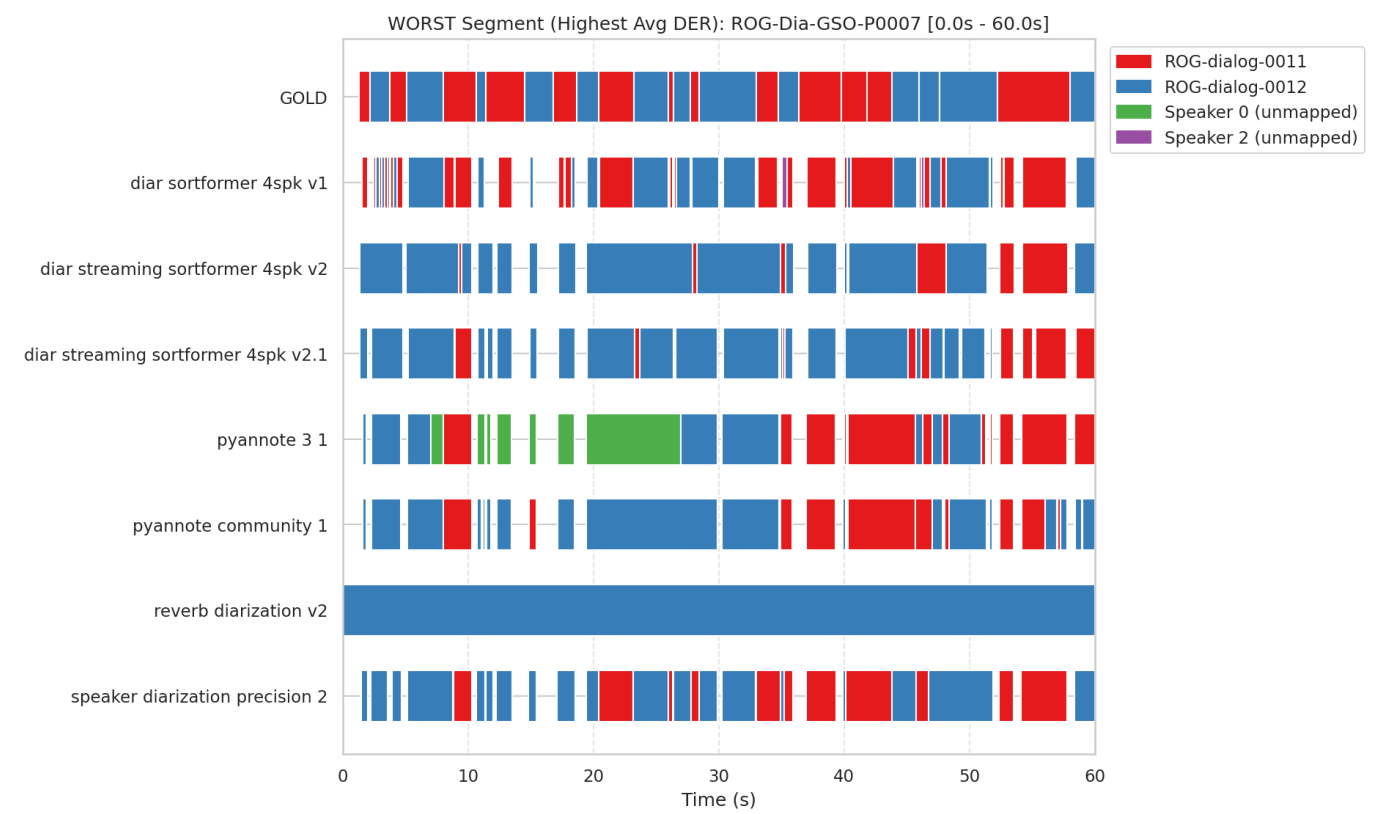
Ena sogovornica je obiskala drugo in v kuhinji za mizo sta opravile pogovor, ki se je snemal.



60-Second Snippets (Zoom-in)

Below are 60-second zoomed-in windows showing where the models performed best and worst (based on average DER).



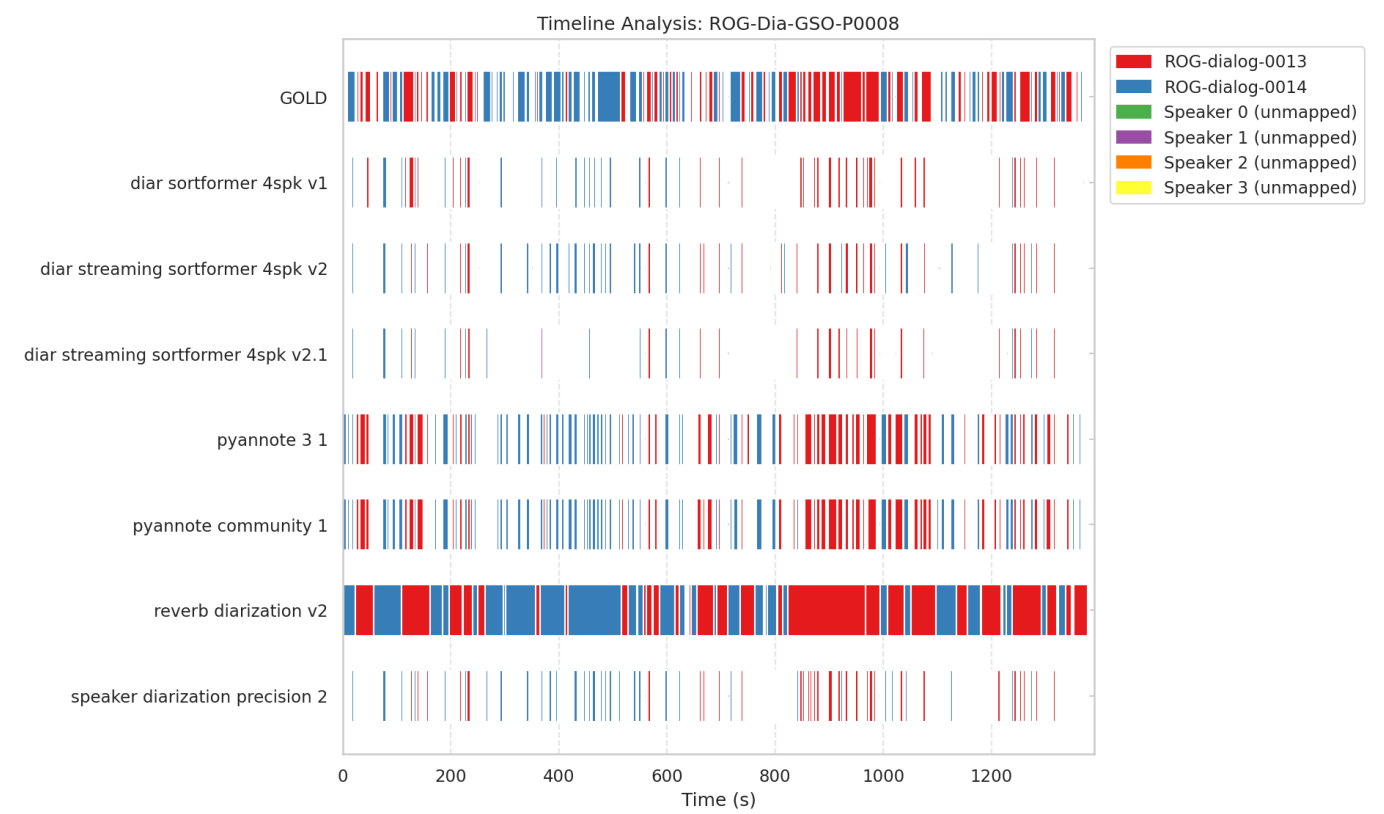


Model	DER	Miss	FA	Conf	Pur	Cov	VRAM (GB)
speaker diarization precision 2	20.64	17.89	1.35	1.41	74.68	77.02	0.0
diar streaming sortformer 4spk v2	21.86	17.64	1.89	2.34	73.42	75.38	1.5
diar streaming sortformer 4spk v2.1	25.74	21.65	1.57	2.52	73.27	75.85	1.5
pyannote community 1	27.16	18.19	2.26	6.71	68.98	70.33	1.6
diar sortformer 4spk v1	28.54	18.15	5.47	4.91	72.34	74.54	104.4
pyannote 3 1	29.76	18.18	2.26	9.31	68.83	68.17	1.6
reverb diarization v2	47.51	18.71	0.26	28.54	52.75	100.00	4.3

File: ROG-Dia-GSO-P0008

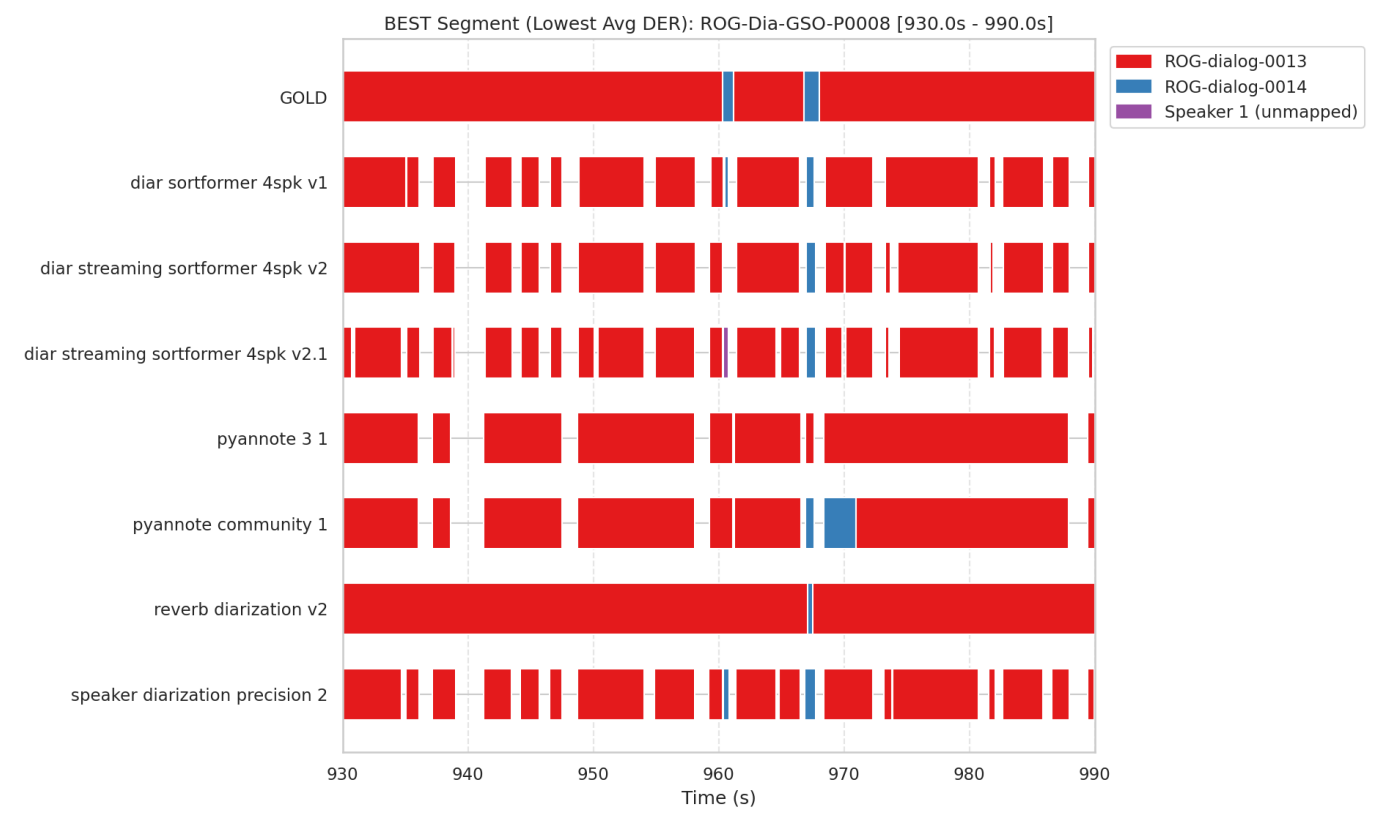
Domain: Diskusija | Quality: 4 | Device: Iphone 13 mini

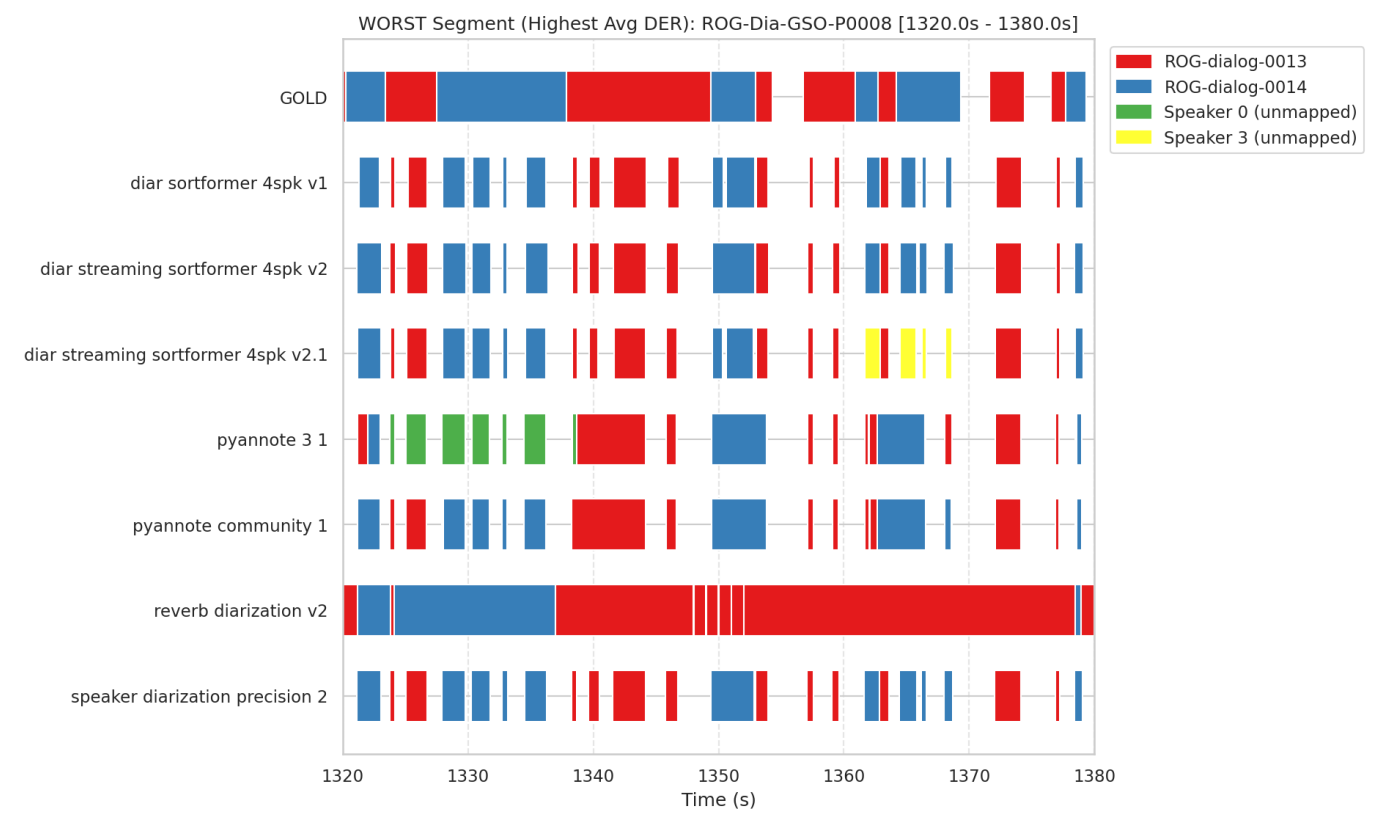
Pogovor med bratoma o kosilu ter o tehnologiji ter argumentiranje prednosti in slabosti.



60-Second Snippets (Zoom-in)

Below are 60-second zoomed-in windows showing where the models performed best and worst (based on average DER).



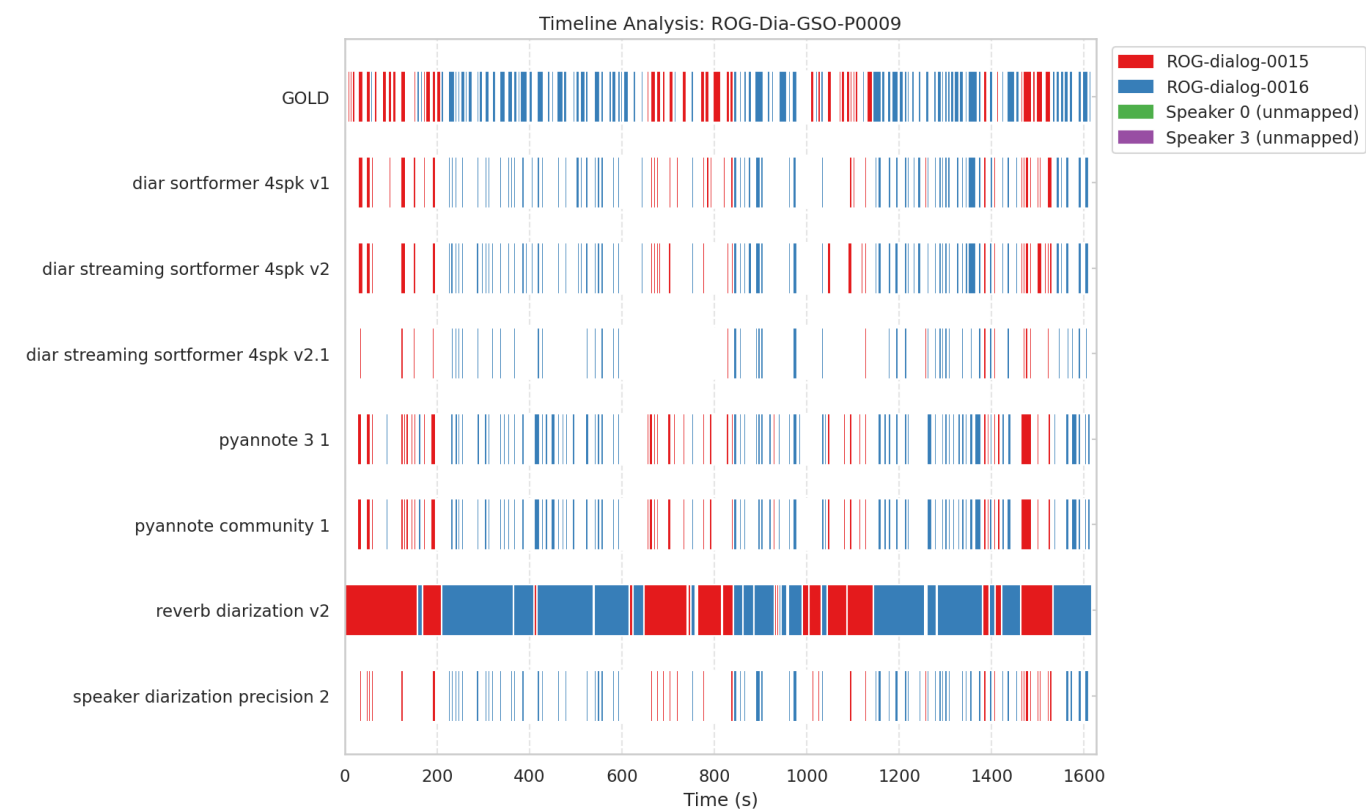


Model	DER	Miss	FA	Conf	Pur	Cov	VRAM (GB)
reverb diarization v2	24.36	2.61	4.14	17.61	79.48	78.63	4.3
speaker diarization precision 2	30.43	28.39	1.26	0.78	94.63	95.81	0.0
pyannote community 1	30.51	21.35	2.79	6.37	87.16	87.72	1.6
pyannote 3 1	31.66	21.35	2.78	7.52	86.64	86.46	1.6
diar streaming sortformer 4spk v2	33.74	31.03	1.61	1.10	94.52	95.12	1.3
diar sortformer 4spk v1	35.09	31.03	2.98	1.08	93.61	94.96	63.9
diar streaming sortformer 4spk v2.1	40.82	34.89	1.53	4.40	95.17	90.34	1.3

File: ROG-Dia-GSO-P0009

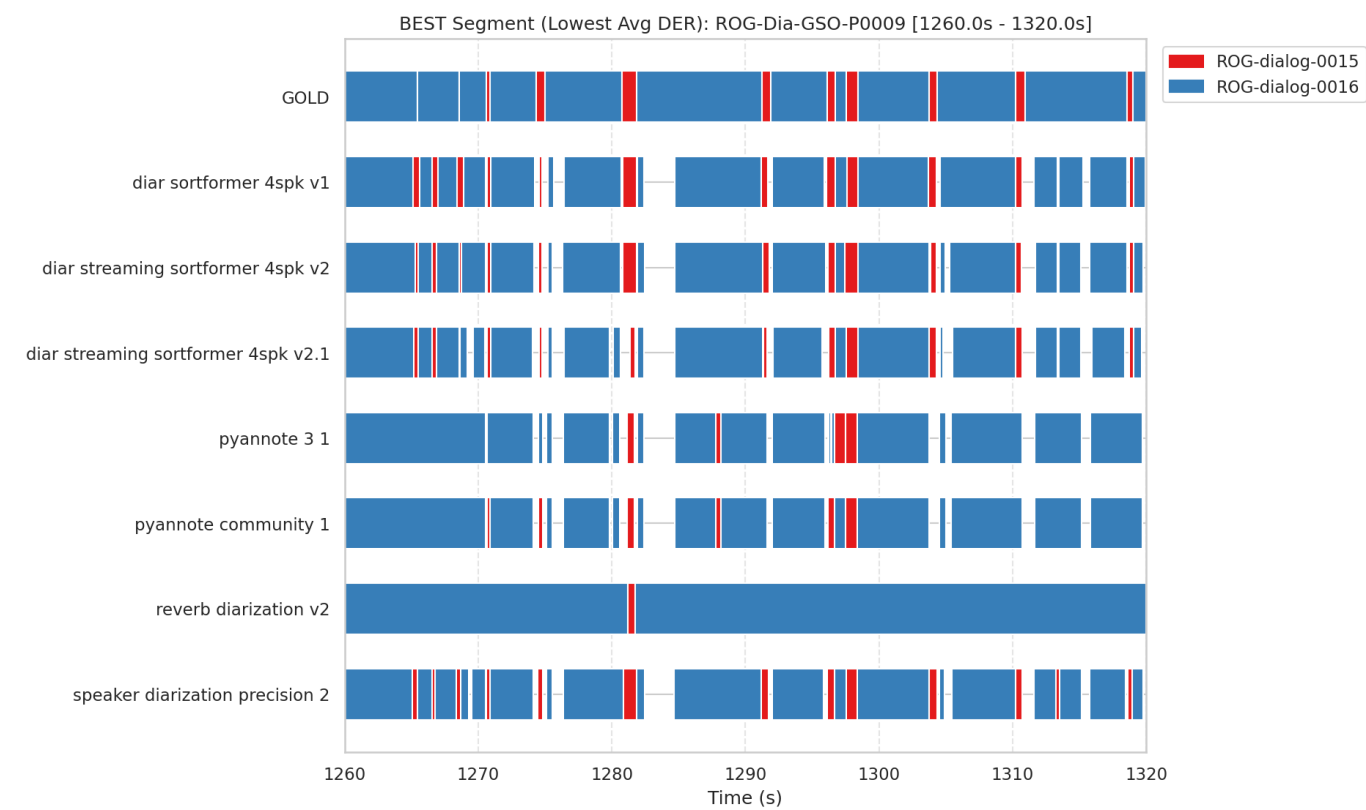
Domain: Družabni pogovor | **Quality:** 5 | **Device:** Mobilni telefon Iphone SE 2020

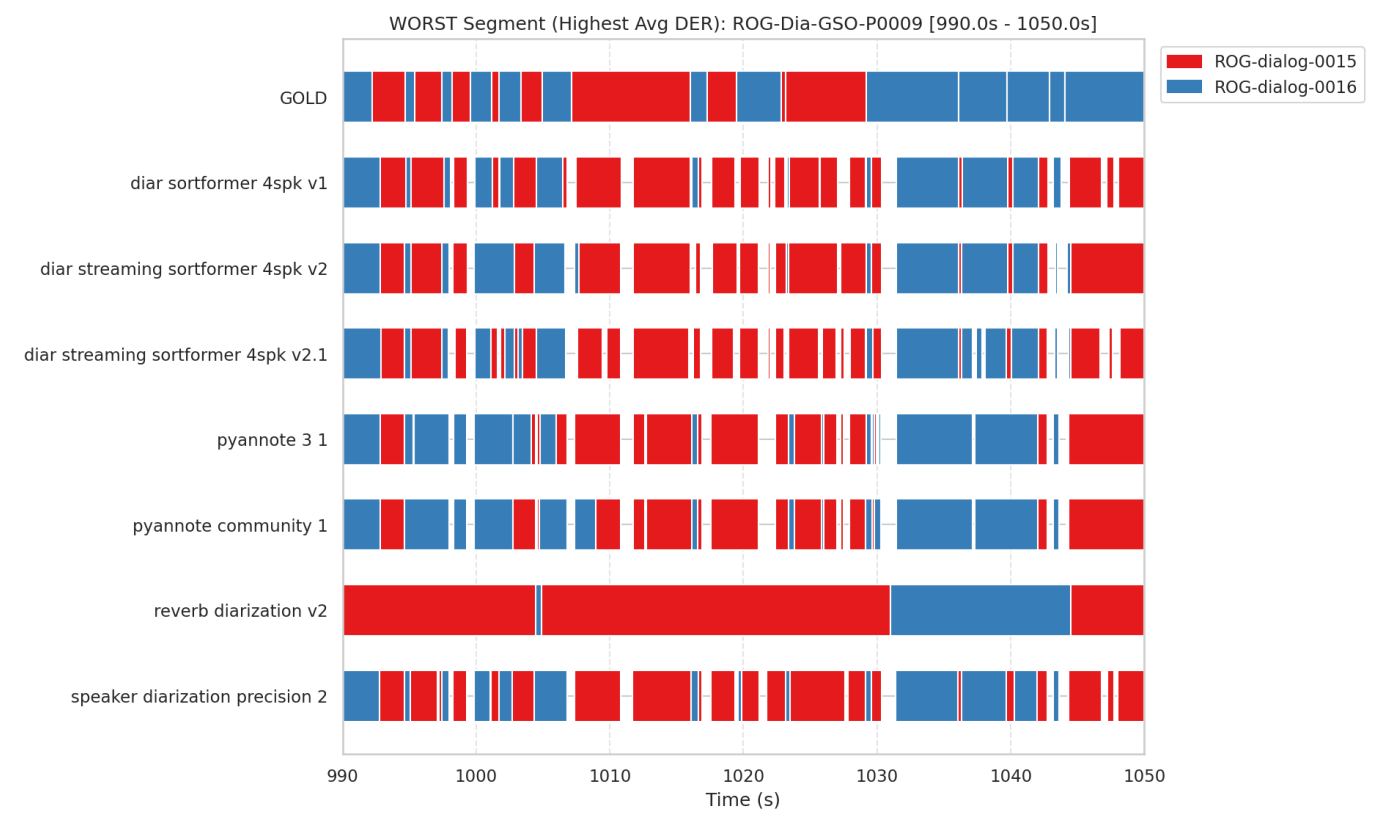
*Posnetek je nastal v domačem okolju enega izmed govorcev med njim in njegovim dolgoletnim sosedom in prijateljem na pobudo snemalke. *



60-Second Snippets (Zoom-in)

Below are 60-second zoomed-in windows showing where the models performed best and worst (based on average DER).



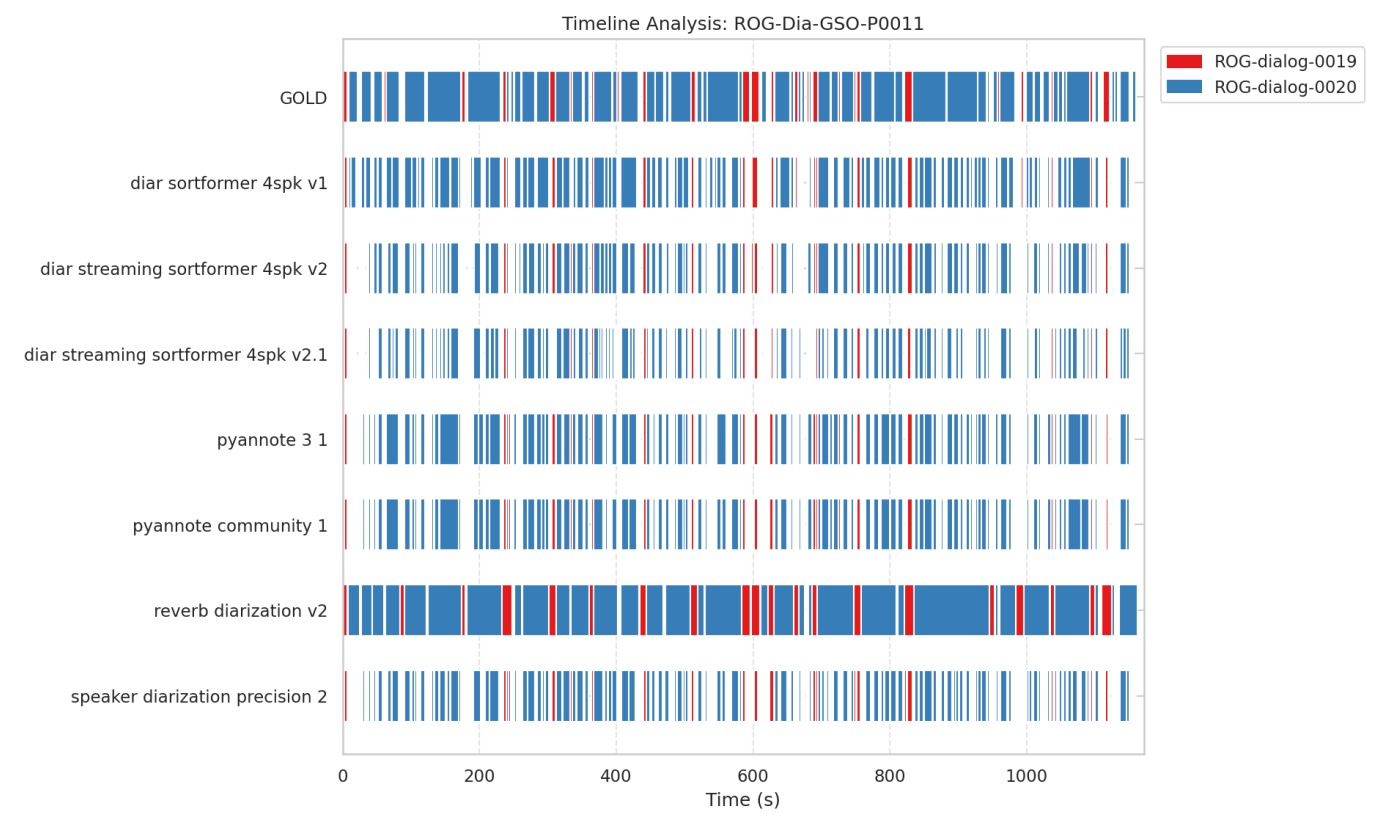


Model	DER	Miss	FA	Conf	Pur	Cov	VRAM (GB)
reverb diarization v2	19.85	12.45	1.51	5.89	81.11	79.43	4.3
diar sortformer 4spk v1	19.95	17.05	2.36	0.54	81.33	84.33	87.1
speaker diarization precision 2	20.13	17.51	2.21	0.41	81.04	83.87	0.0
diar streaming sortformer 4spk v2	21.49	19.35	1.64	0.50	82.00	85.00	1.4
pyannote 3 1	24.34	19.46	1.81	3.06	79.43	80.25	1.6
pyannote community 1	24.40	19.46	1.81	3.13	79.36	80.35	1.6
diar streaming sortformer 4spk v2.1	25.68	23.51	1.59	0.59	81.58	85.24	1.4

File: ROG-Dia-GSO-P0011

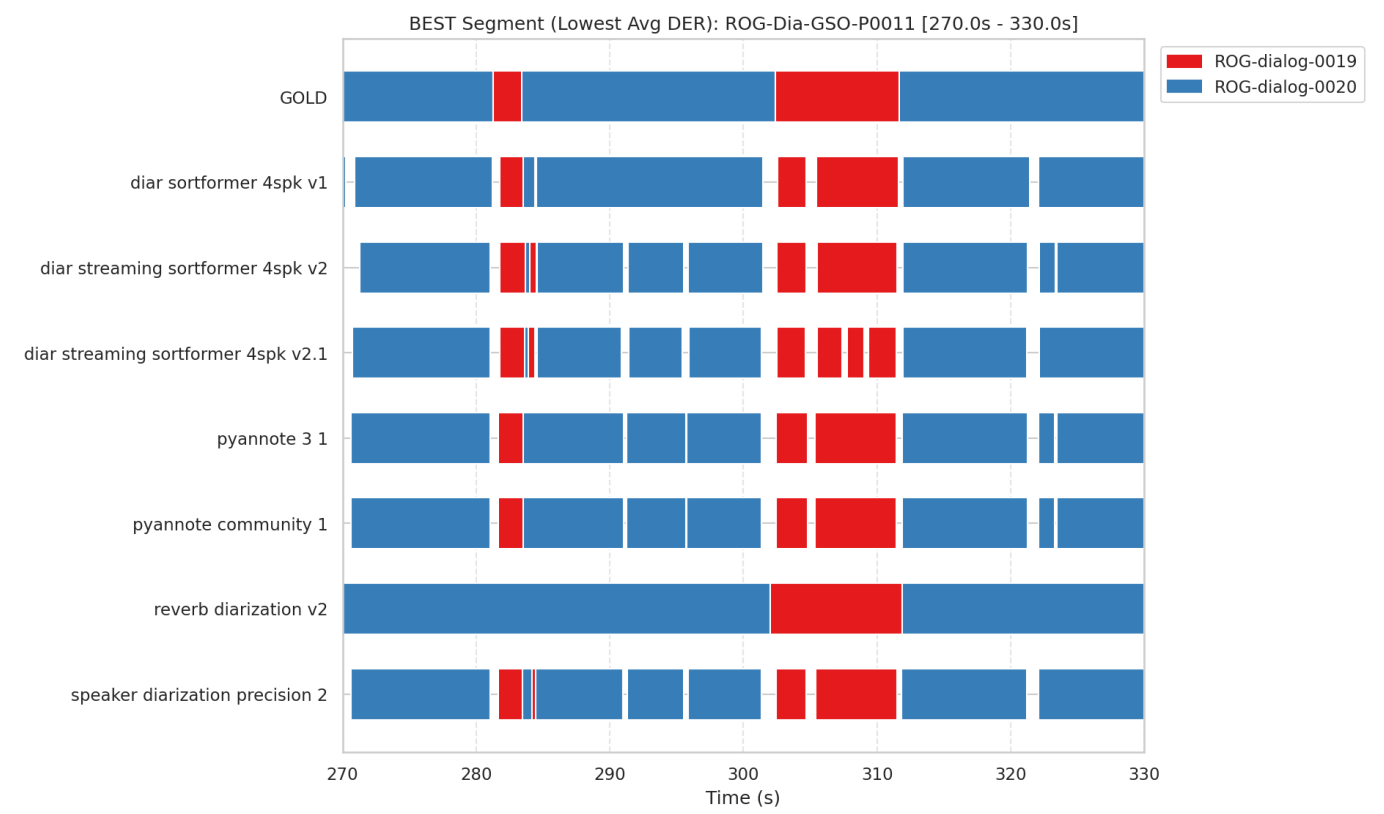
Domain: Intervju | **Quality:** 4 | **Device:** Mobilna naprava Iphone 13, aplikacija Voice Record.

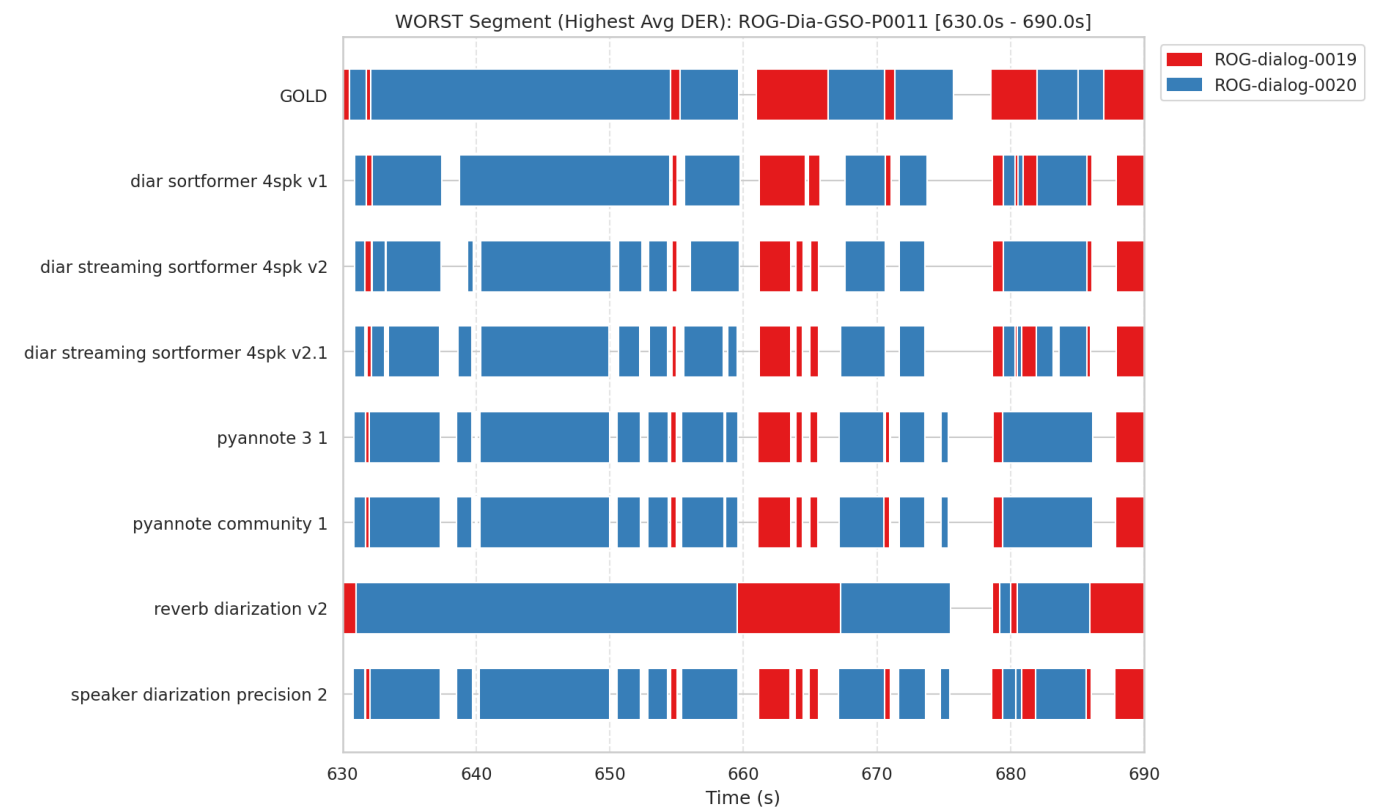
Govorca sta ženska in moški, oba stara okrog 50 let. Je večer, sedita za kuhinjsko mizo. Ženska postavlja moškemu vprašanja o domačem kraju, običajih in spominih iz otroštva ter mladosti. Moški odgovarja na njena vprašanja, ona pa ga med tem tudi spomni na pomembne točke, ki bi jih lahko vključil v svoje odgovore.



60-Second Snippets (Zoom-in)

Below are 60-second zoomed-in windows showing where the models performed best and worst (based on average DER).



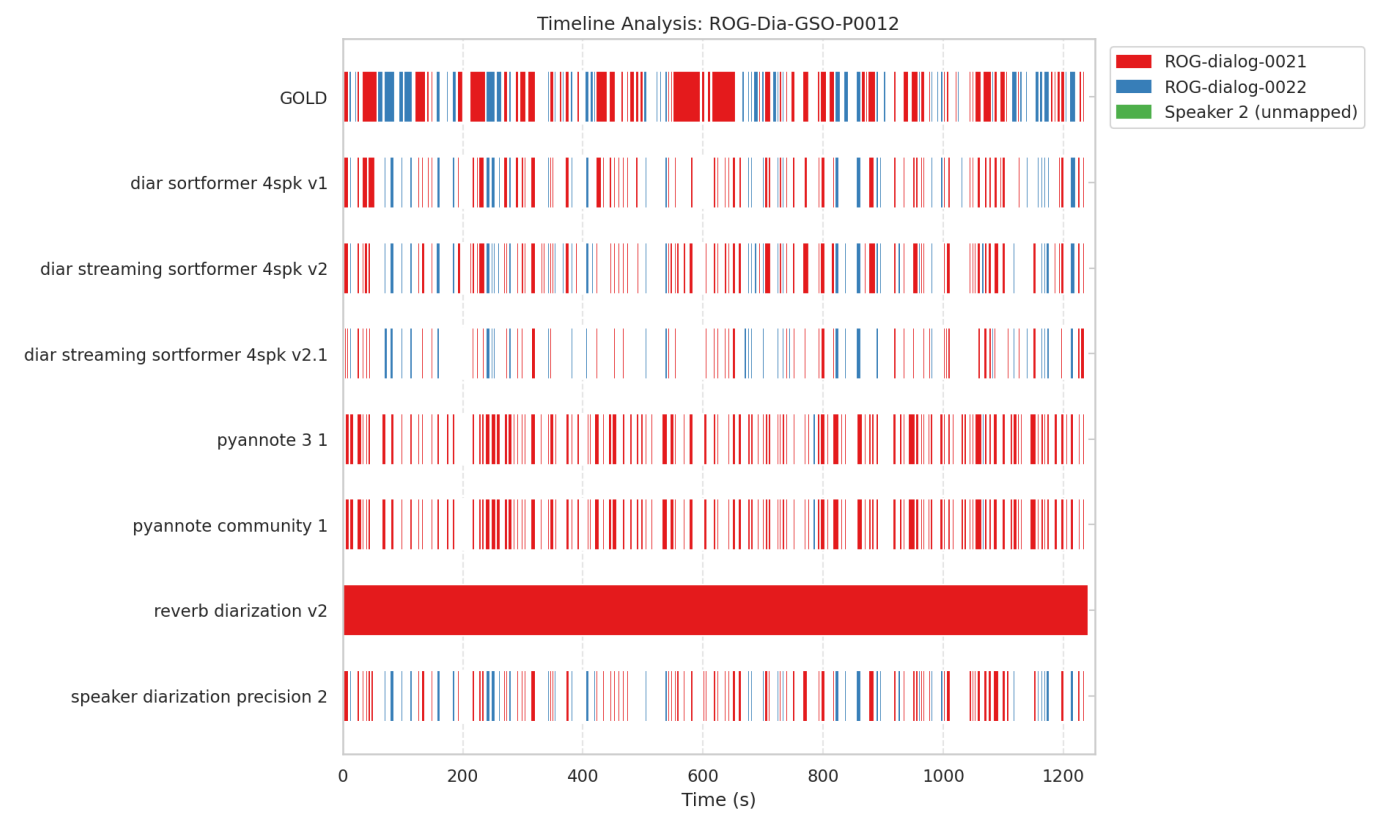


Model	DER	Miss	FA	Conf	Pur	Cov	VRAM (GB)
reverb diarization v2	12.40	4.24	4.27	3.90	91.30	90.96	4.3
diar sortformer 4spk v1	13.38	12.26	0.93	0.19	92.84	94.38	45.7
speaker diarization precision 2	15.44	14.34	0.98	0.12	92.41	94.09	0.0
pyannote 3 1	15.62	14.15	1.06	0.41	92.28	93.58	1.6
pyannote community 1	15.73	14.15	1.06	0.53	92.16	93.51	1.6
diar streaming sortformer 4spk v2	18.50	17.47	0.90	0.13	92.65	94.20	1.1
diar streaming sortformer 4spk v2.1	20.17	19.18	0.88	0.11	92.40	94.33	1.1

File: ROG-Dia-GSO-P0012

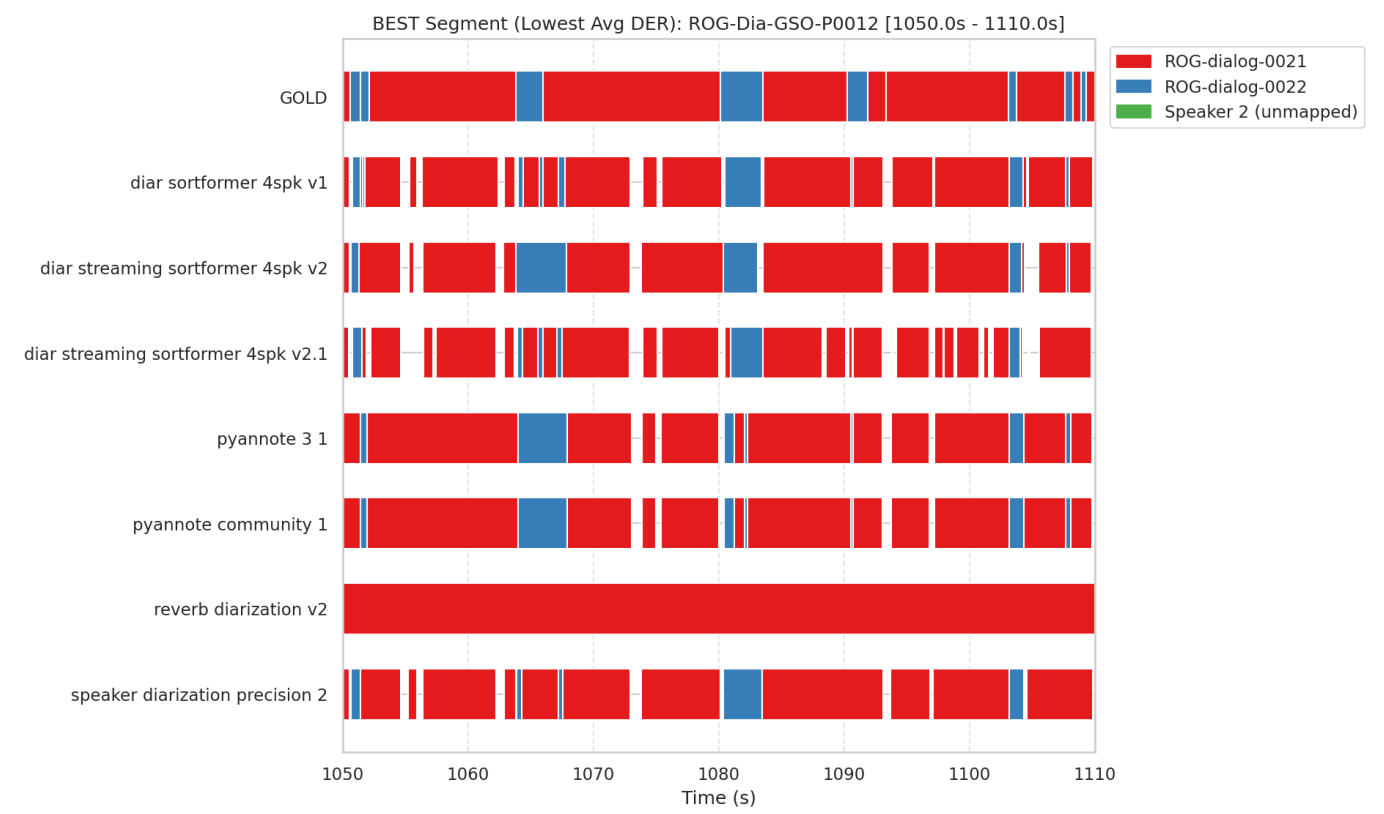
Domain: Intervju | Quality: 4 | Device: Pametni telefon Samsung Galaxy a53

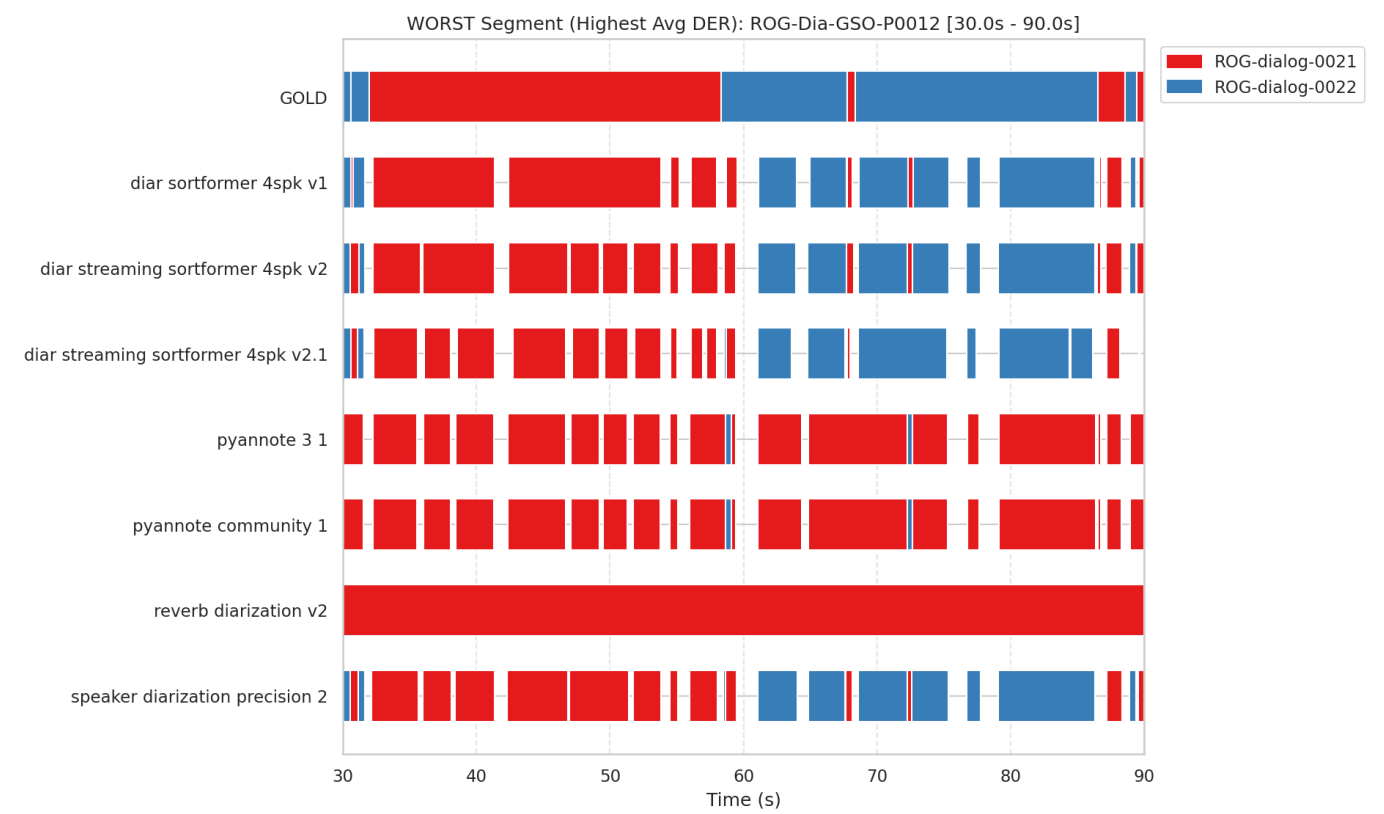
Ena od sogovornic je prišla na obisk k drugi. Usedli sta se za mizo in pričeli s pogovorom.



60-Second Snippets (Zoom-in)

Below are 60-second zoomed-in windows showing where the models performed best and worst (based on average DER).





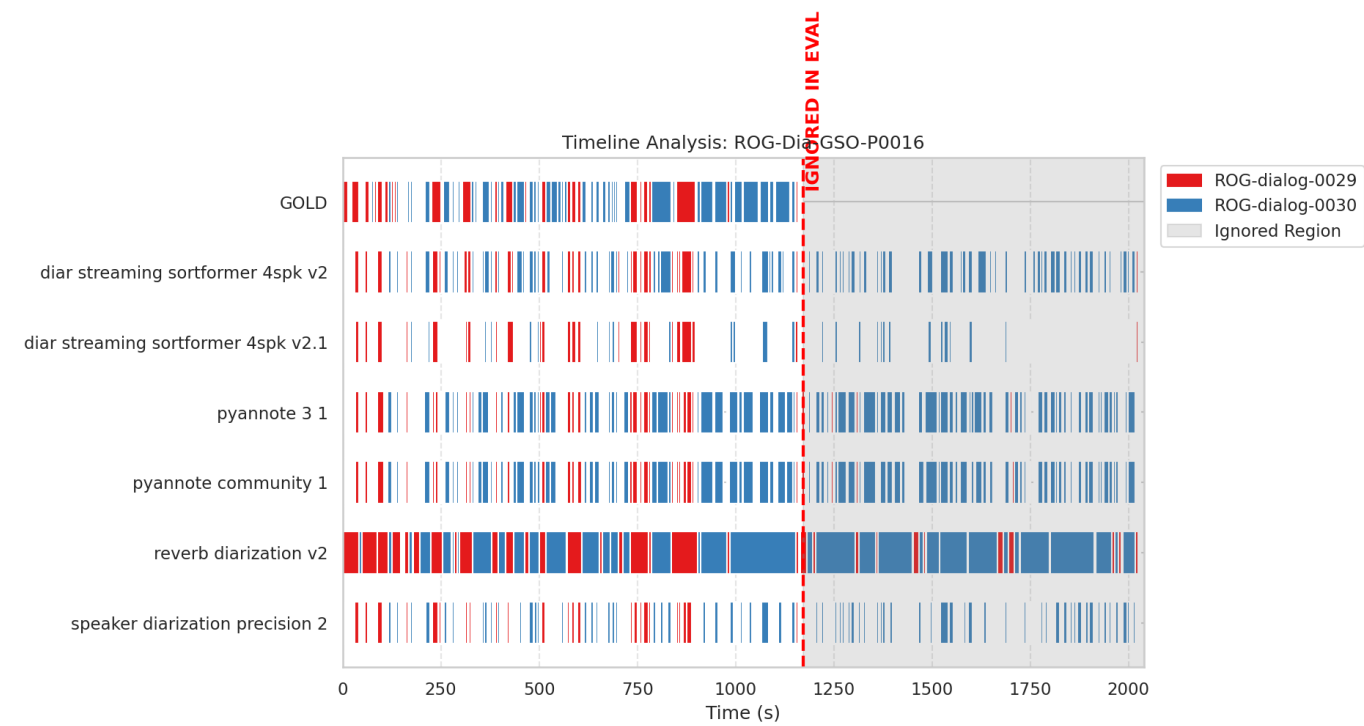
Model	DER	Miss	FA	Conf	Pur	Cov	VRAM (GB)
speaker diarization precision 2	16.33	14.67	1.20	0.46	84.69	87.01	0.0
diar streaming sortformer 4spk v2	16.44	14.27	1.54	0.63	84.09	85.88	1.2
diar sortformer 4spk v1	18.02	16.07	1.34	0.61	84.59	87.79	51.9
diar streaming sortformer 4spk v2.1	24.51	22.92	0.84	0.76	85.20	88.36	1.2
reverb diarization v2	40.45	9.65	0.76	30.04	60.31	100.00	4.3
pyannote community 1	42.31	15.32	1.51	25.47	59.17	90.25	1.6
pyannote 3 1	42.31	15.32	1.51	25.47	59.17	90.25	1.6

File: ROG-Dia-GSO-P0016

Domain: Diskusija | Quality: 3 | Device: pametni telefon Huawei P30 PRO

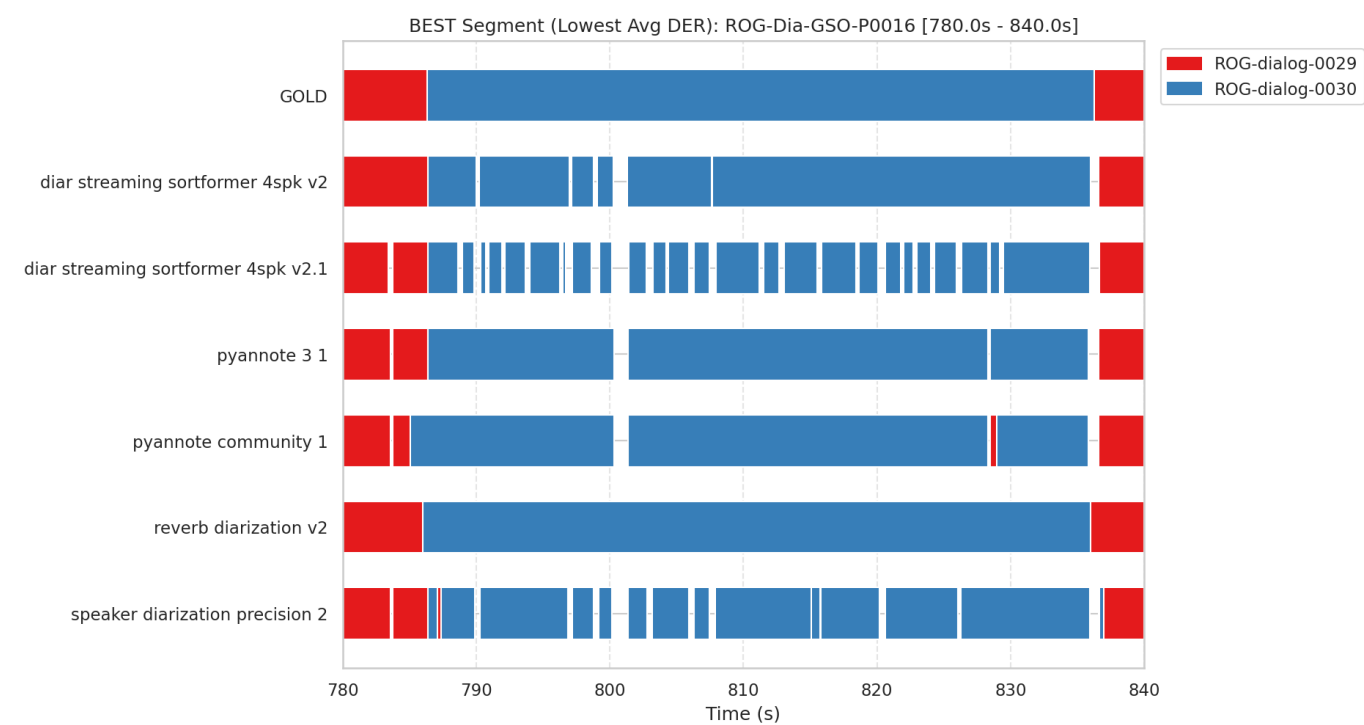
*Starejši moški in ženska imata diskusijo o svojem otroštvu, odraščanju na kmetih. *

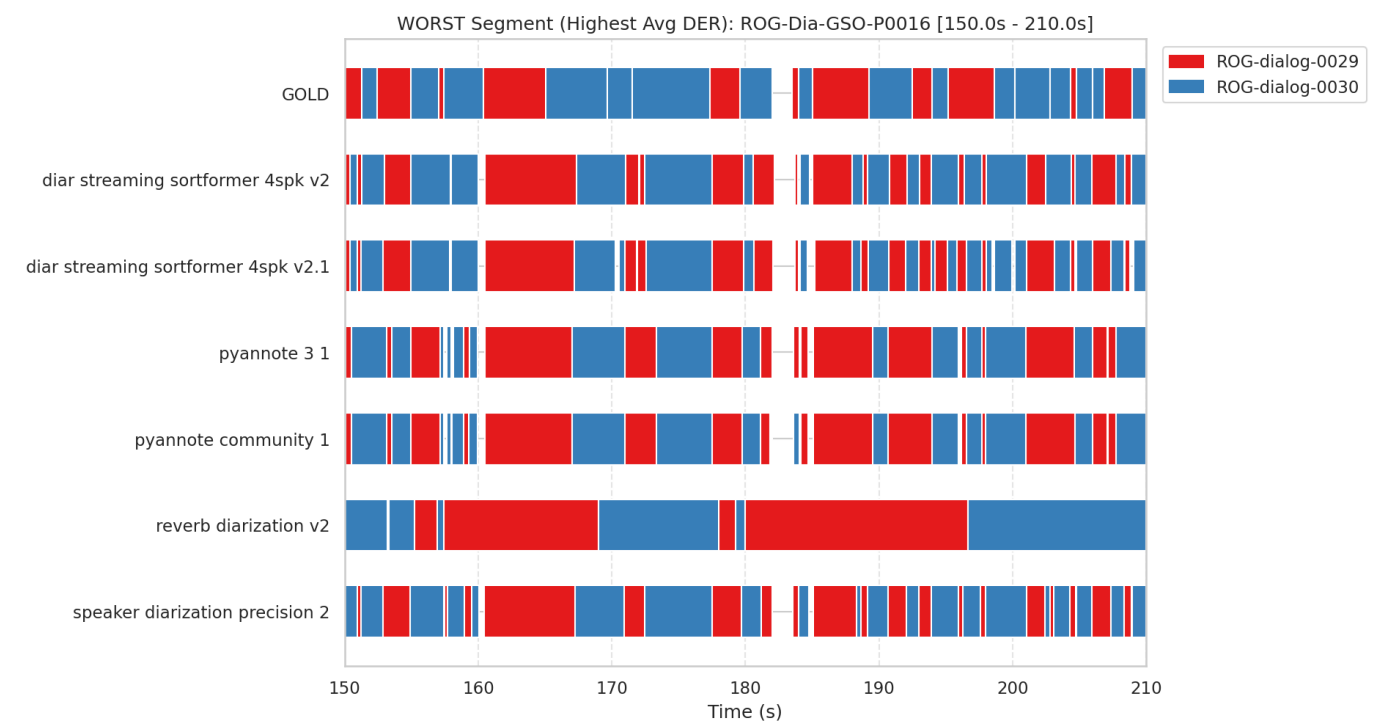
⚠ **ERRATA APPLIED:** Evaluation bounded to 1172.092s.



60-Second Snippets (Zoom-in)

Below are 60-second zoomed-in windows showing where the models performed best and worst (based on average DER).



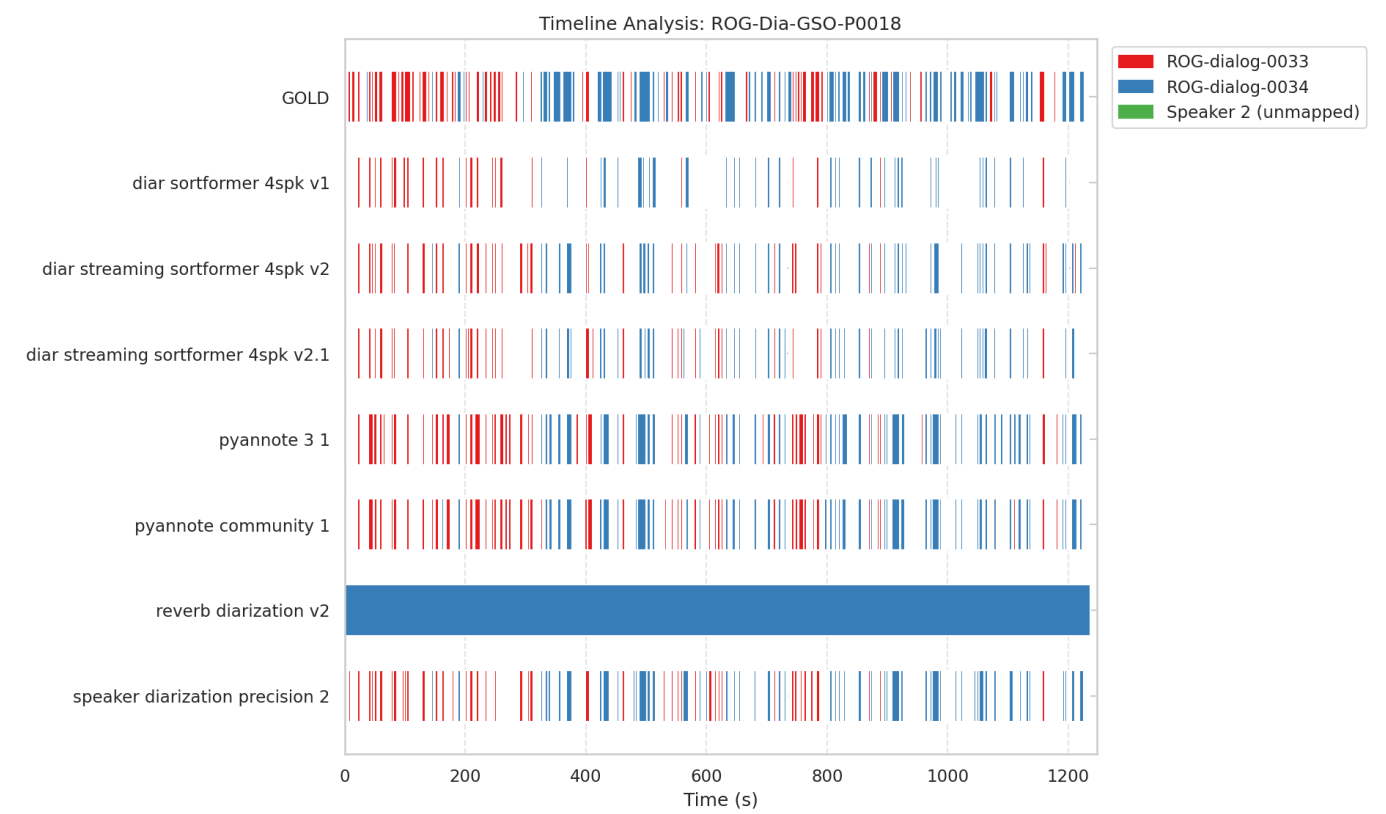


Model	DER	Miss	FA	Conf	Pur	Cov	VRAM (GB)	Status
diar streaming sortformer 4spk v2	13.43	11.58	1.36	0.49	81.18	83.06	1.6	nan
speaker diarization precision 2	15.05	13.23	1.39	0.42	80.94	83.05	0.0	nan
pyannote 3 1	15.07	11.60	1.66	1.81	80.45	81.95	1.6	nan
pyannote community 1	15.39	11.60	1.66	2.12	80.16	81.67	1.6	nan
reverb diarization v2	17.20	11.91	0.88	4.41	82.87	81.94	4.3	nan
diar streaming sortformer 4spk v2.1	19.27	17.57	1.21	0.50	81.25	83.67	1.6	nan
diar sortformer 4spk v1	nan	nan	nan	nan	nan	nan	nan	OOM/ERR

File: ROG-Dia-GSO-P0018

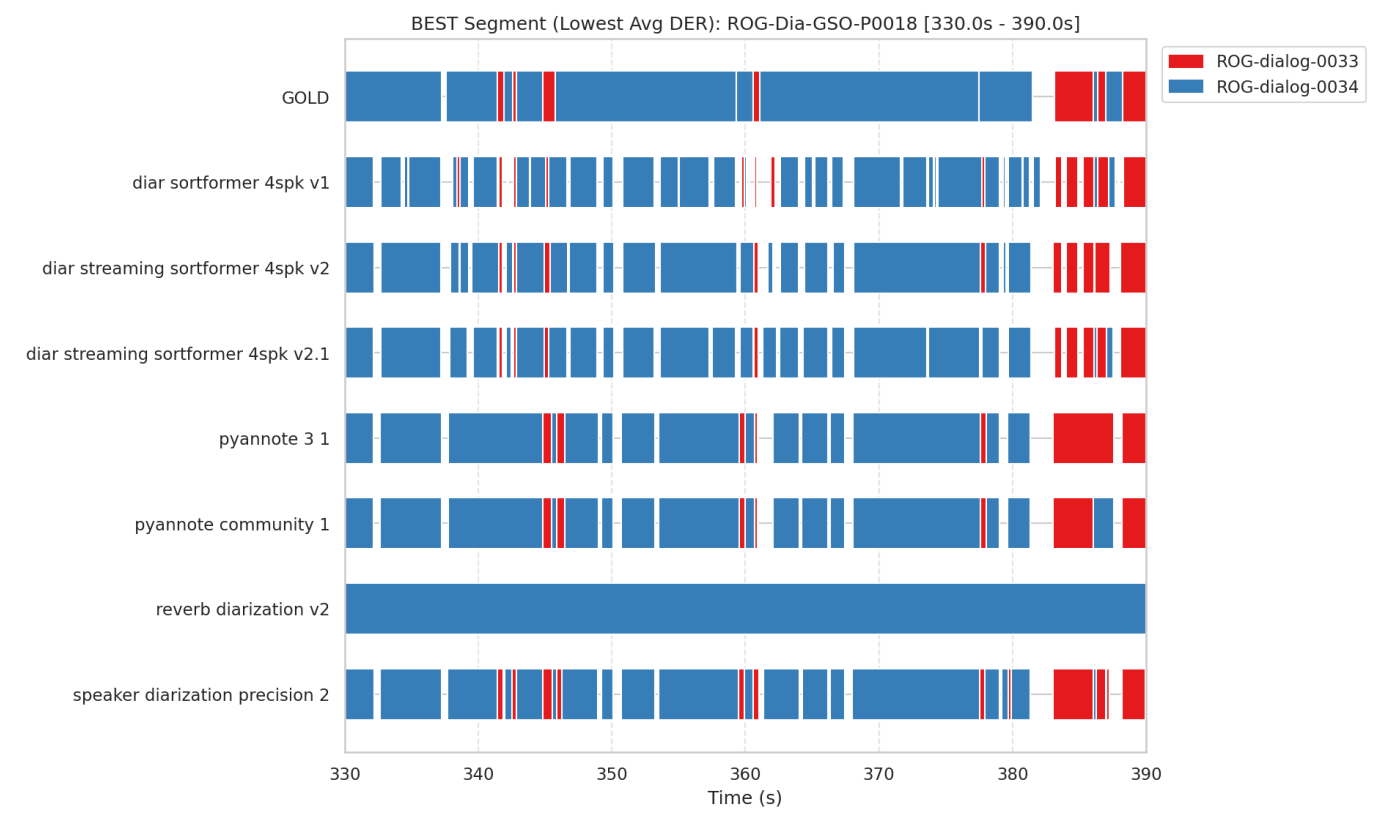
Domain: Družabni pogovor | Quality: 5 | Device: telefon Samsung A52

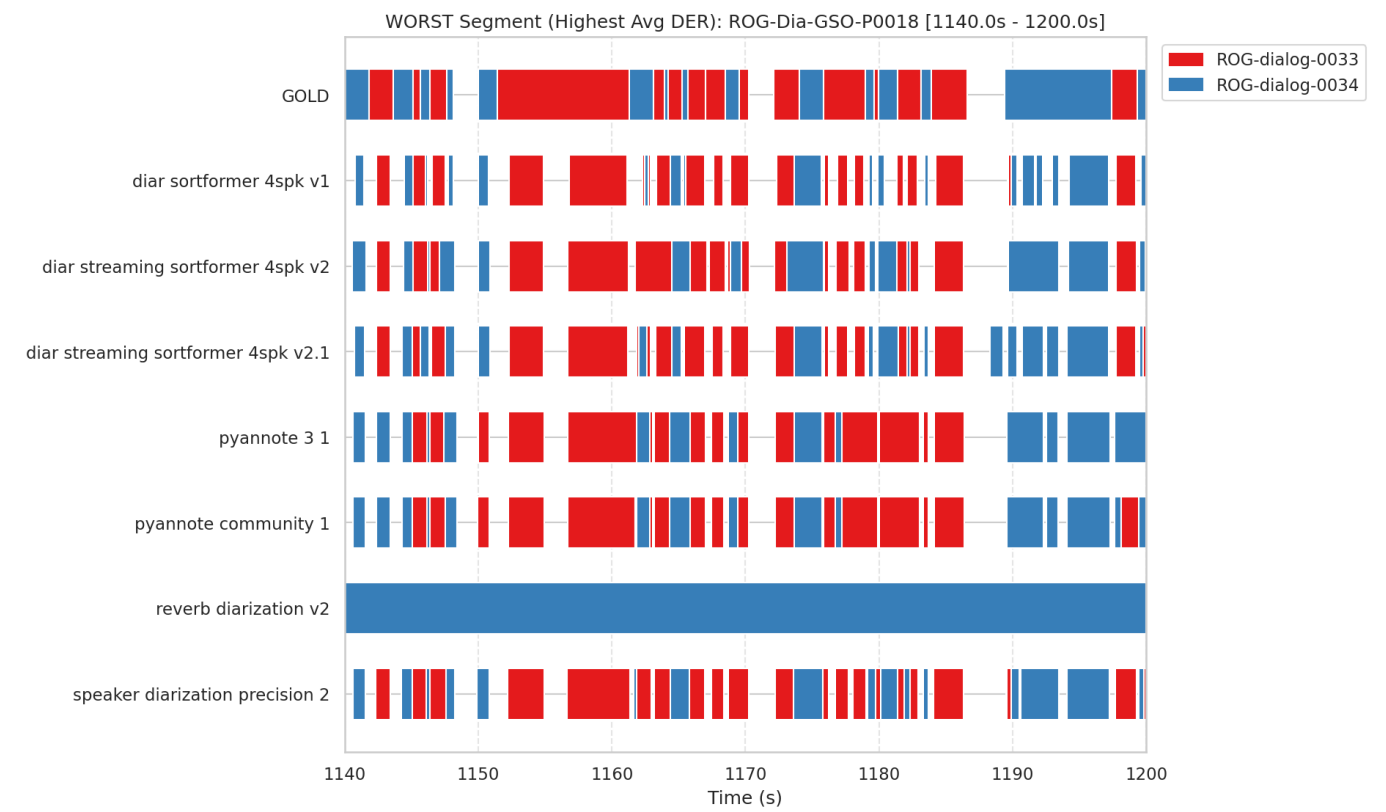
Dobili smo se v župnišču, se pogovorili o prostoru snemanja, zahtevah posnetka. Govorki sta se odločili za spontan družabni pogovor brez vnaprej pripravljenih tem in vprašanj.



60-Second Snippets (Zoom-in)

Below are 60-second zoomed-in windows showing where the models performed best and worst (based on average DER).



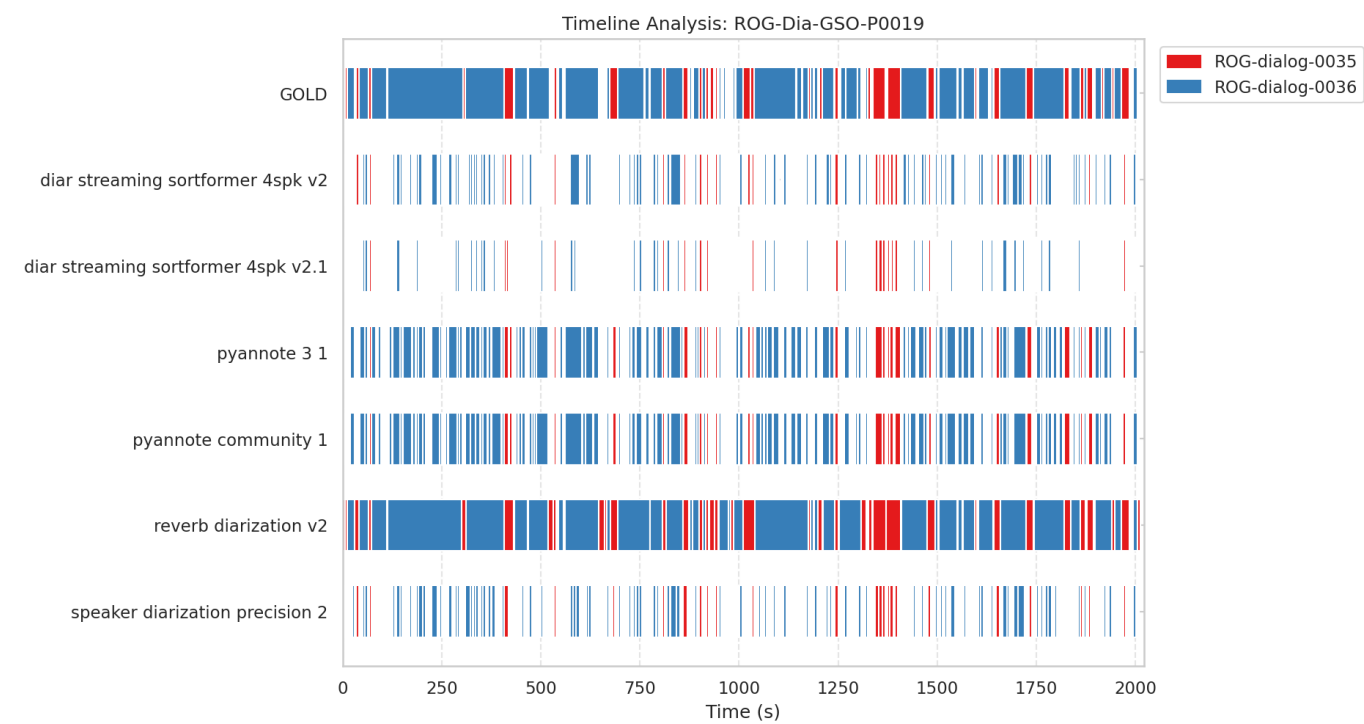


Model	DER	Miss	FA	Conf	Pur	Cov	VRAM (GB)
speaker diarization precision 2	15.50	12.66	2.03	0.81	81.69	83.30	0.0
diar streaming sortformer 4spk v2	20.42	16.66	2.42	1.34	80.64	83.00	1.2
pyannote 3 1	20.89	13.32	2.62	4.95	77.66	78.68	1.6
diar streaming sortformer 4spk v2.1	21.71	17.60	2.82	1.30	80.84	83.79	1.2
pyannote community 1	22.82	13.32	2.62	6.89	75.82	77.08	1.6
diar sortformer 4spk v1	27.26	22.70	2.93	1.63	79.69	82.40	51.6
reverb diarization v2	50.52	11.09	2.84	36.59	52.31	100.00	4.3

File: ROG-Dia-GSO-P0019

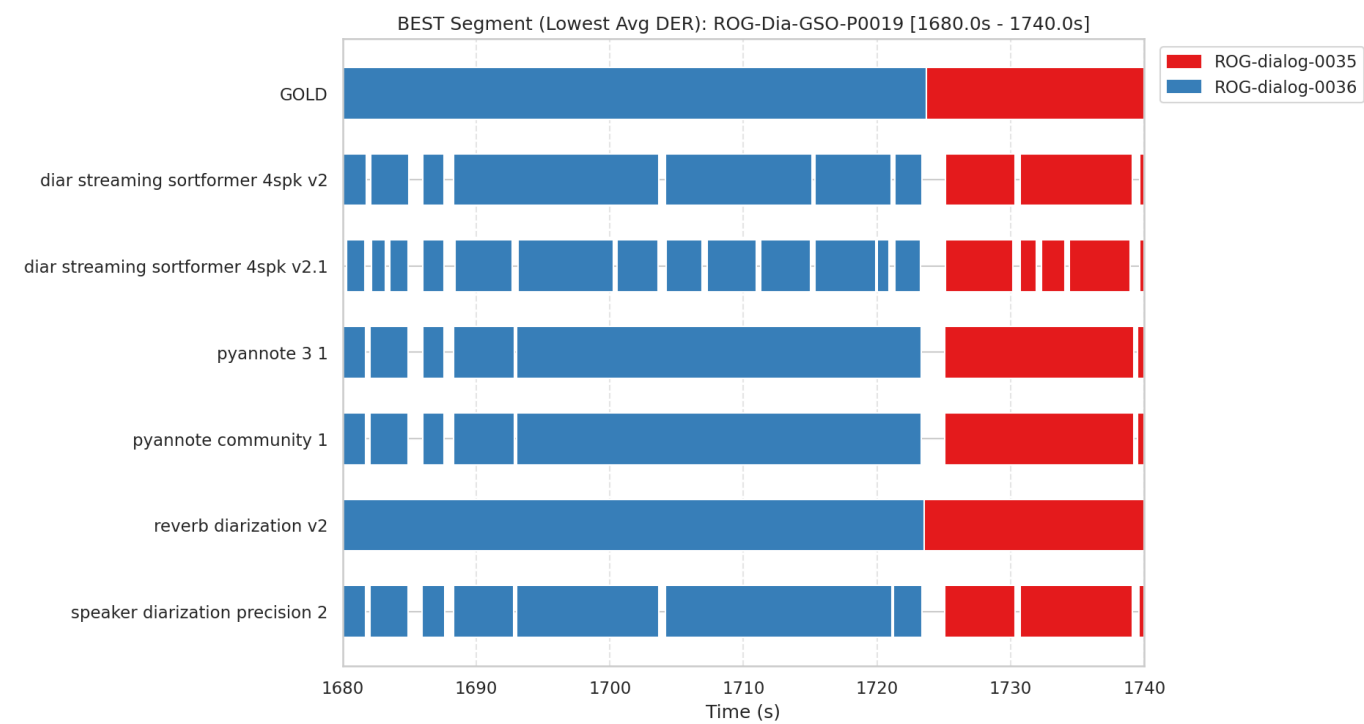
Domain: Pripoved | Quality: 4 | Device: Iphone 13 mini

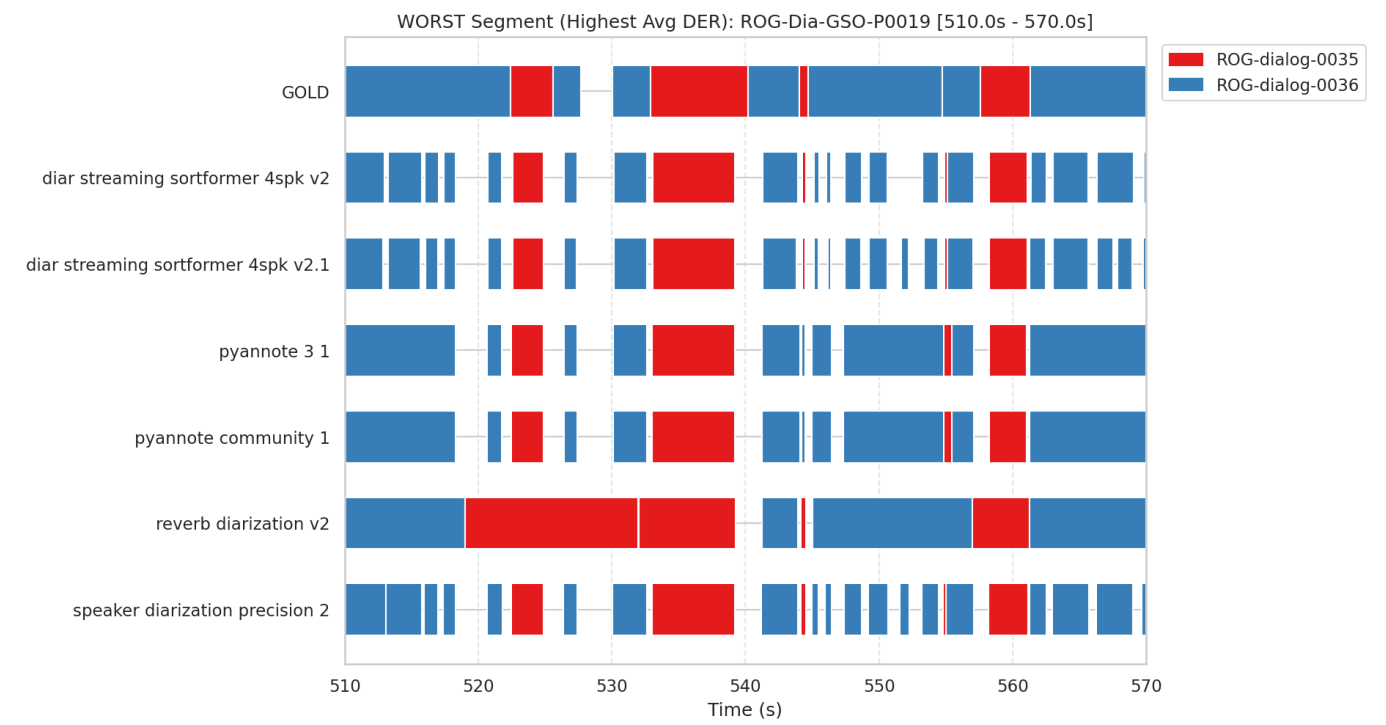
Pogovor med sestro in bratom, ki sta se pogovarjala o potovanjih. Sestra je delila svojo izkušnjo potovanja v tujino, medtem ko jo je brat spraševal še o drugih vidikih potovanja – od priprav pred odhodom do krajev, ki jih je obiskala.



60-Second Snippets (Zoom-in)

Below are 60-second zoomed-in windows showing where the models performed best and worst (based on average DER).



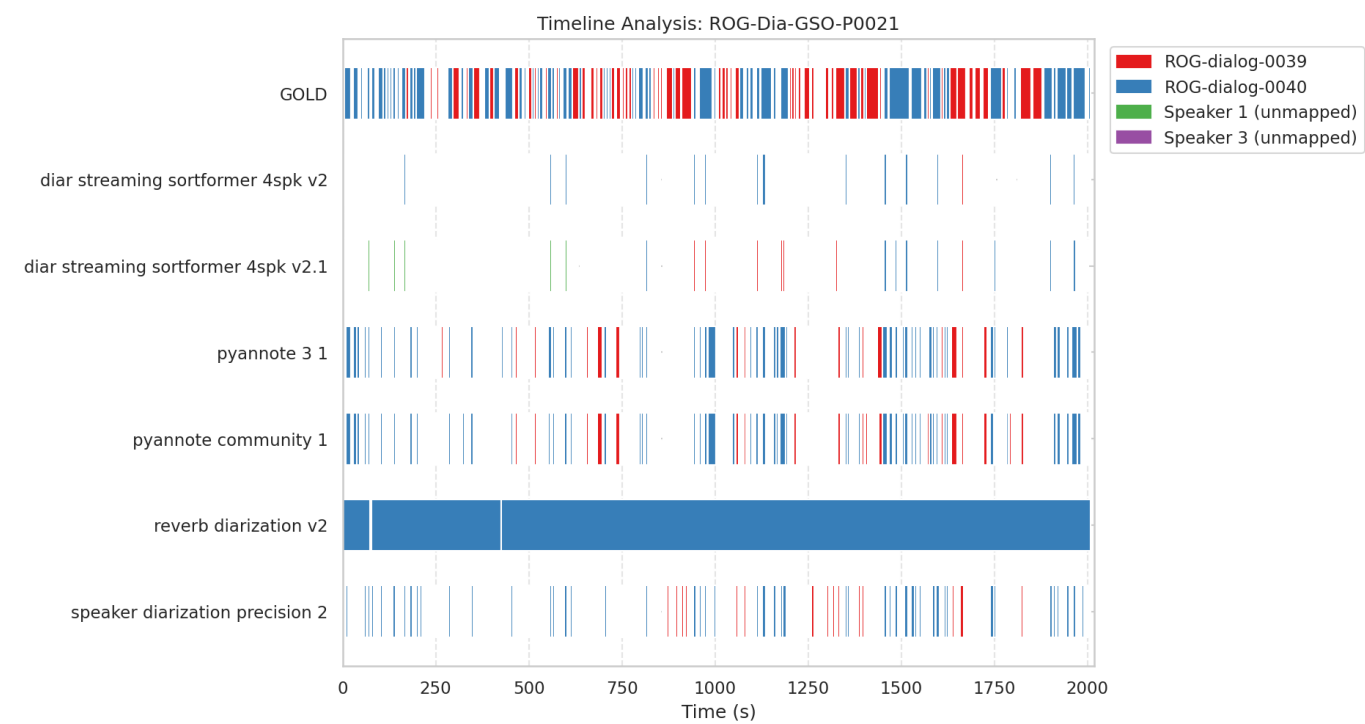


Model	DER	Miss	FA	Conf	Pur	Cov	VRAM (GB)	Status
reverb diarization v2	7.31	1.03	2.58	3.69	94.84	94.93	4.3	nan
pyannote 3 1	12.42	11.98	0.39	0.05	98.59	98.90	1.6	nan
pyannote community 1	12.45	11.98	0.39	0.08	98.55	98.88	1.6	nan
speaker diarization precision 2	17.37	17.20	0.17	0.00	98.95	99.27	0.0	nan
diar streaming sortformer 4spk v2	19.63	19.40	0.22	0.02	98.87	99.22	1.6	nan
diar streaming sortformer 4spk v2.1	23.43	23.27	0.16	0.00	98.94	99.29	1.6	nan
diar sortformer 4spk v1	nan	nan	nan	nan	nan	nan	nan	OOM/ERR

File: ROG-Dia-GSO-P0021

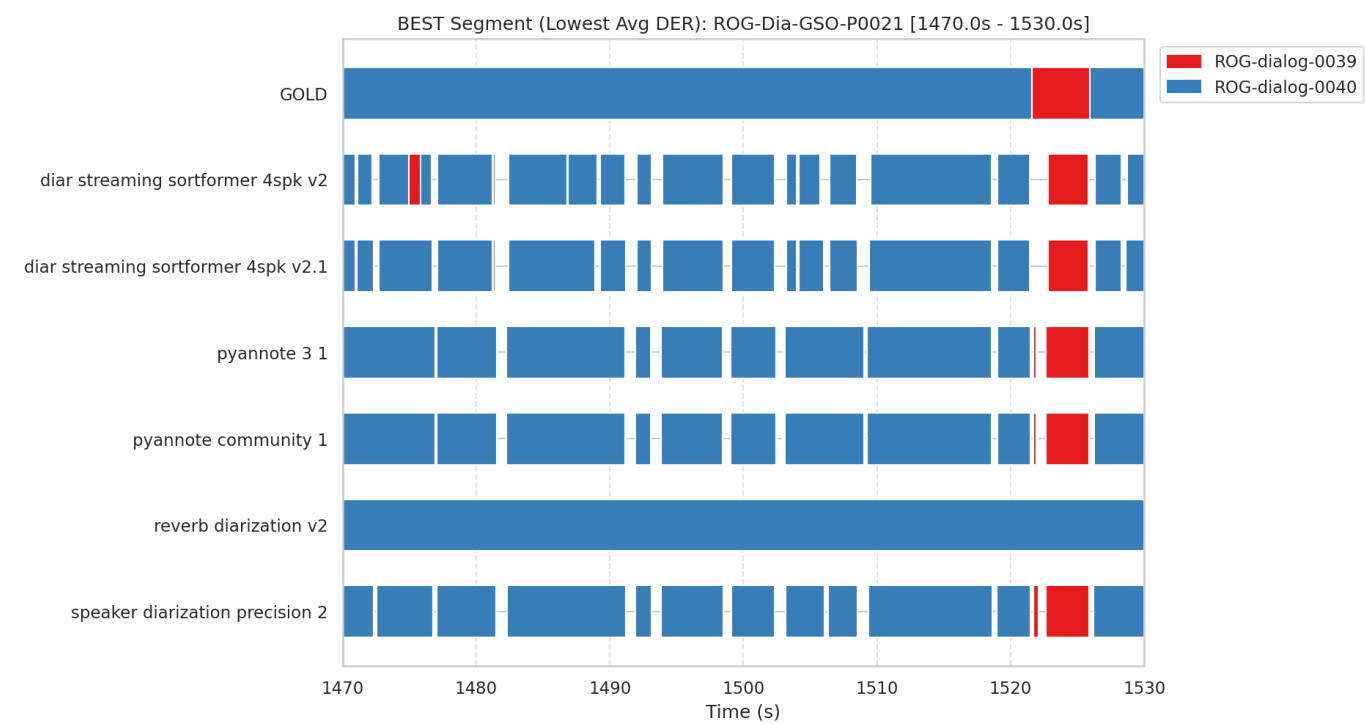
Domain: Družabni pogovor | Quality: 4 | Device: Apple iPhone 15 Pro

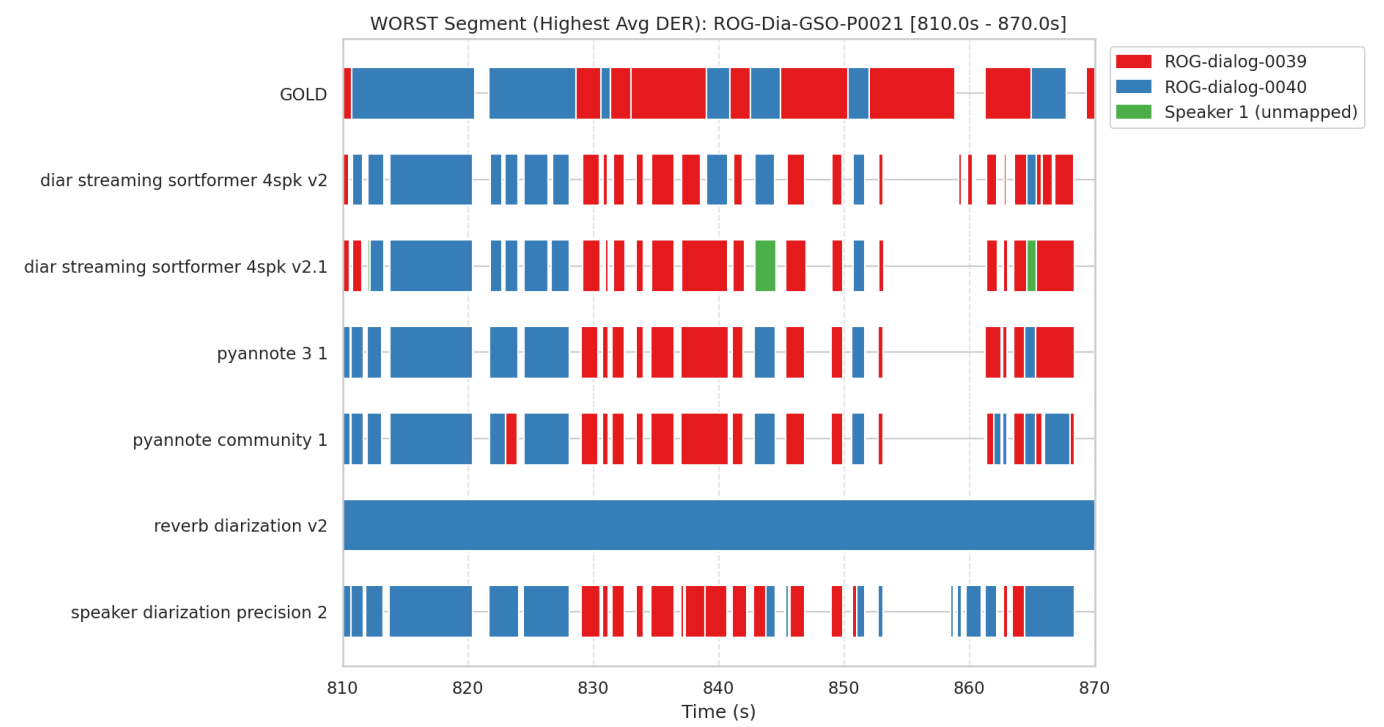
Prijateljski pogovor med 2 osebama, ki se pogovarjata o splošnih temah; delo, prosti čas, hišni ljubljenci, načrti za prihodnost, opis dela, itd



60-Second Snippets (Zoom-in)

Below are 60-second zoomed-in windows showing where the models performed best and worst (based on average DER).



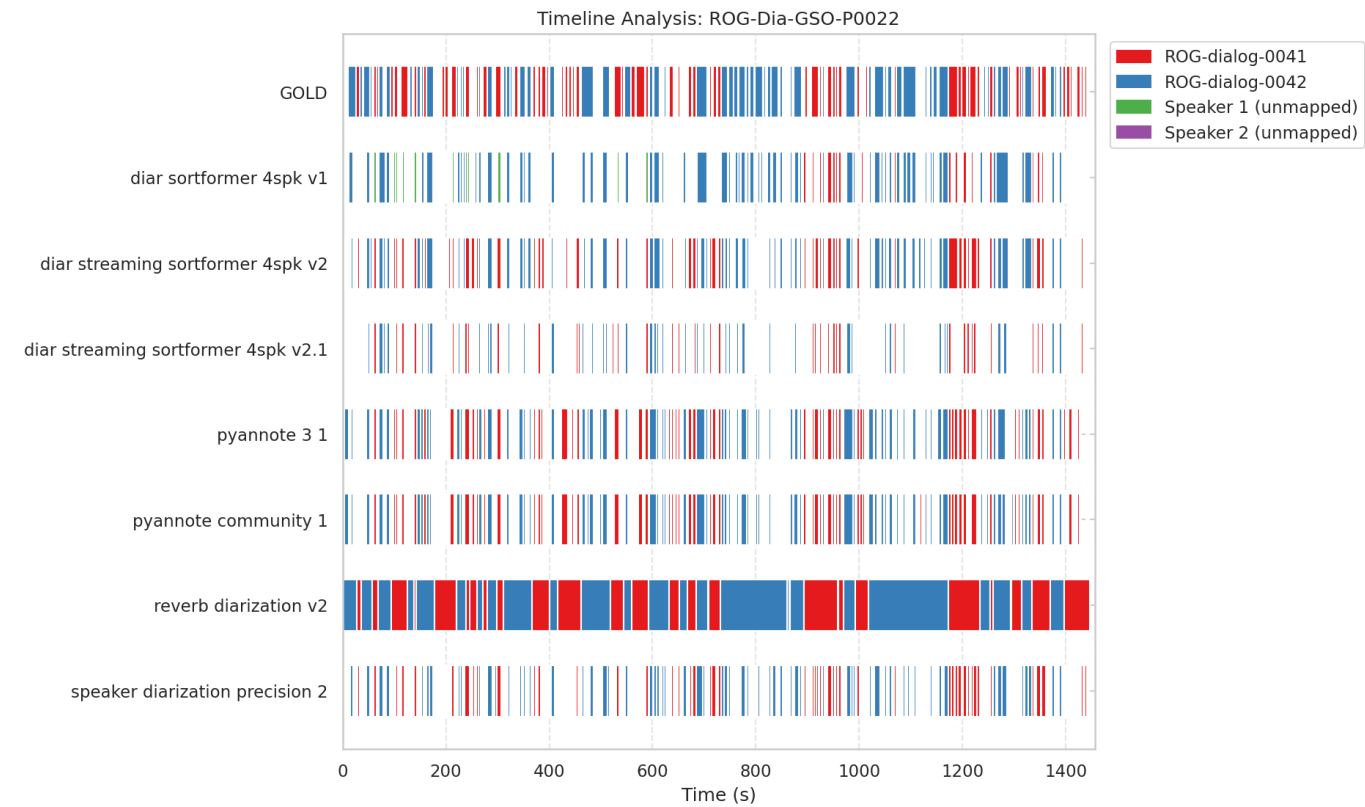


Model	DER	Miss	FA	Conf	Pur	Cov	VRAM (GB)	Status
speaker diarization precision 2	23.13	20.43	1.18	1.52	93.42	94.67	0.0	nan
pyannote 3 1	25.47	19.47	1.07	4.94	89.92	90.62	1.6	nan
pyannote community 1	26.32	19.47	1.07	5.78	88.92	90.03	1.6	nan
diar streaming sortformer 4spk v2	30.98	28.54	0.87	1.57	93.25	94.44	1.6	nan
reverb diarization v2	45.89	2.36	1.72	41.81	55.89	100.00	4.3	nan
diar streaming sortformer 4spk v2.1	60.22	27.44	1.32	31.46	82.09	54.38	1.6	nan
diar sortformer 4spk v1	nan	nan	nan	nan	nan	nan	nan	OOM/ERR

File: ROG-Dia-GSO-P0022

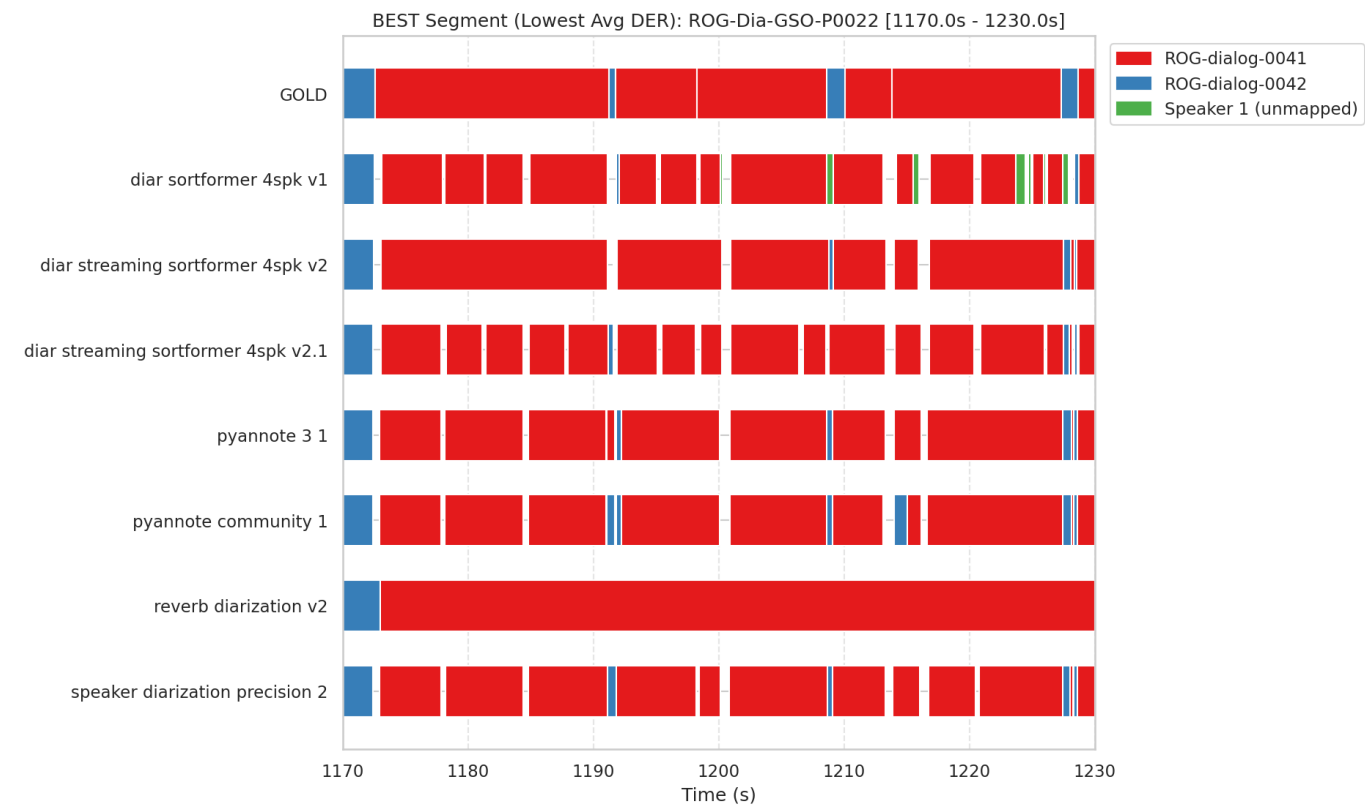
Domain: Družabni pogovor | **Quality:** 5 | **Device:** Pametni telefon iphone 11 z aplikacijo Voice Record Pro z naročenimi nastavitvami.

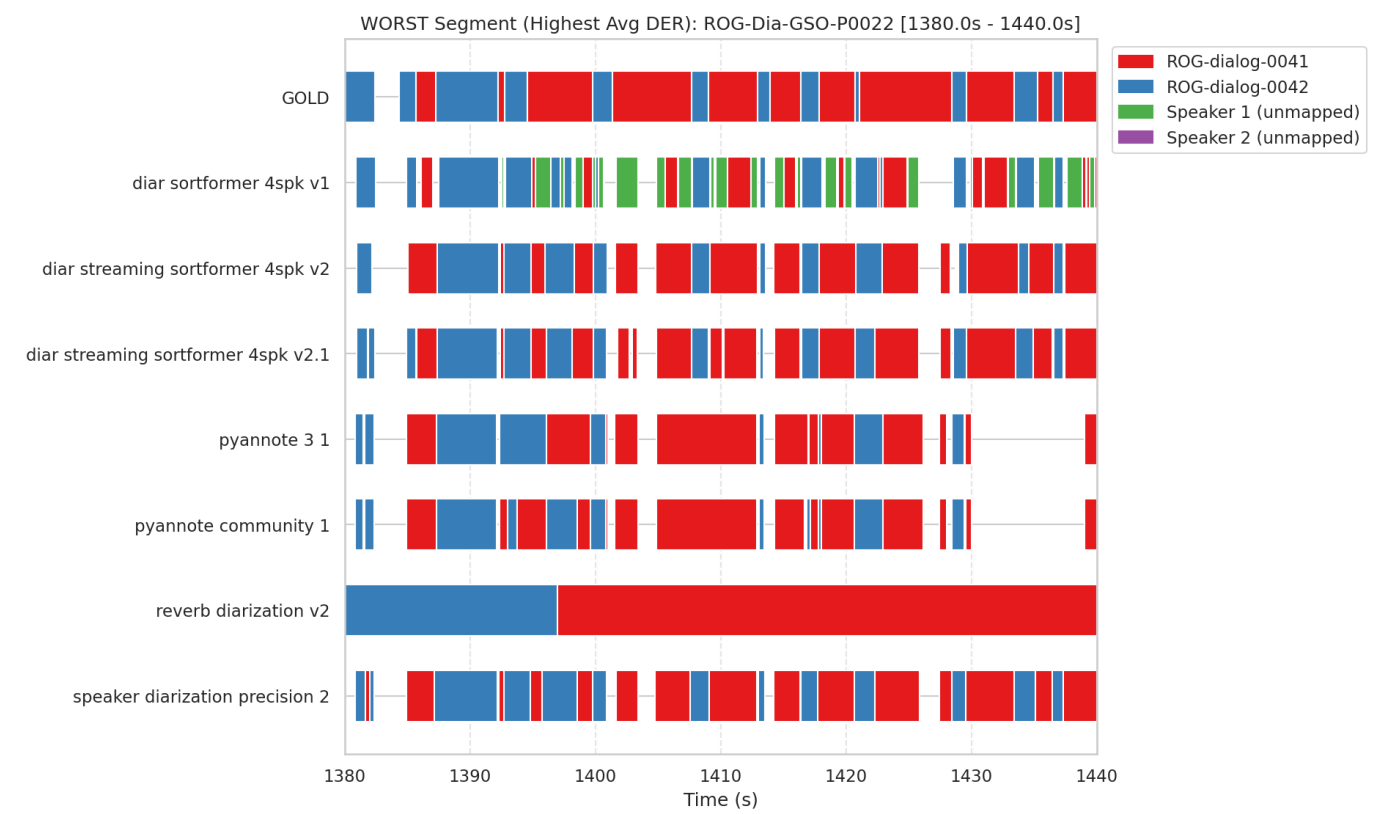
Pogovor med sorodnikoma o glasbenih skupinah in glasbenih izkušnjah.



60-Second Snippets (Zoom-in)

Below are 60-second zoomed-in windows showing where the models performed best and worst (based on average DER).



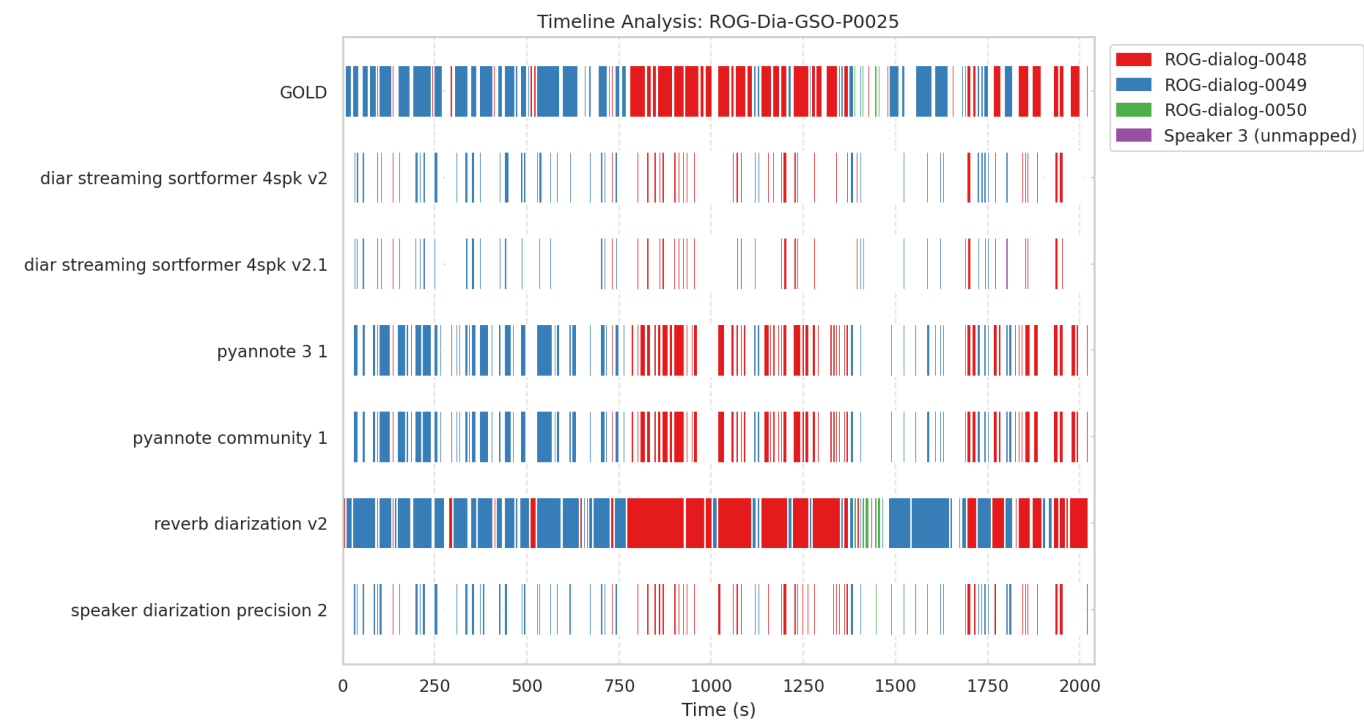


Model	DER	Miss	FA	Conf	Pur	Cov	VRAM (GB)
speaker diarization precision 2	13.01	12.16	0.76	0.09	85.81	87.56	0.0
diar streaming sortformer 4spk v2	16.38	15.60	0.58	0.20	86.73	88.65	1.3
pyannote 3 1	16.77	13.53	1.07	2.17	84.24	84.79	1.6
pyannote community 1	16.99	13.53	1.07	2.39	84.02	84.69	1.6
reverb diarization v2	18.21	8.92	0.95	8.34	82.42	80.18	4.3
diar streaming sortformer 4spk v2.1	19.12	18.40	0.64	0.08	86.10	88.85	1.3
diar sortformer 4spk v1	35.48	17.05	0.92	17.51	87.11	71.72	69.9

File: ROG-Dia-GSO-P0025

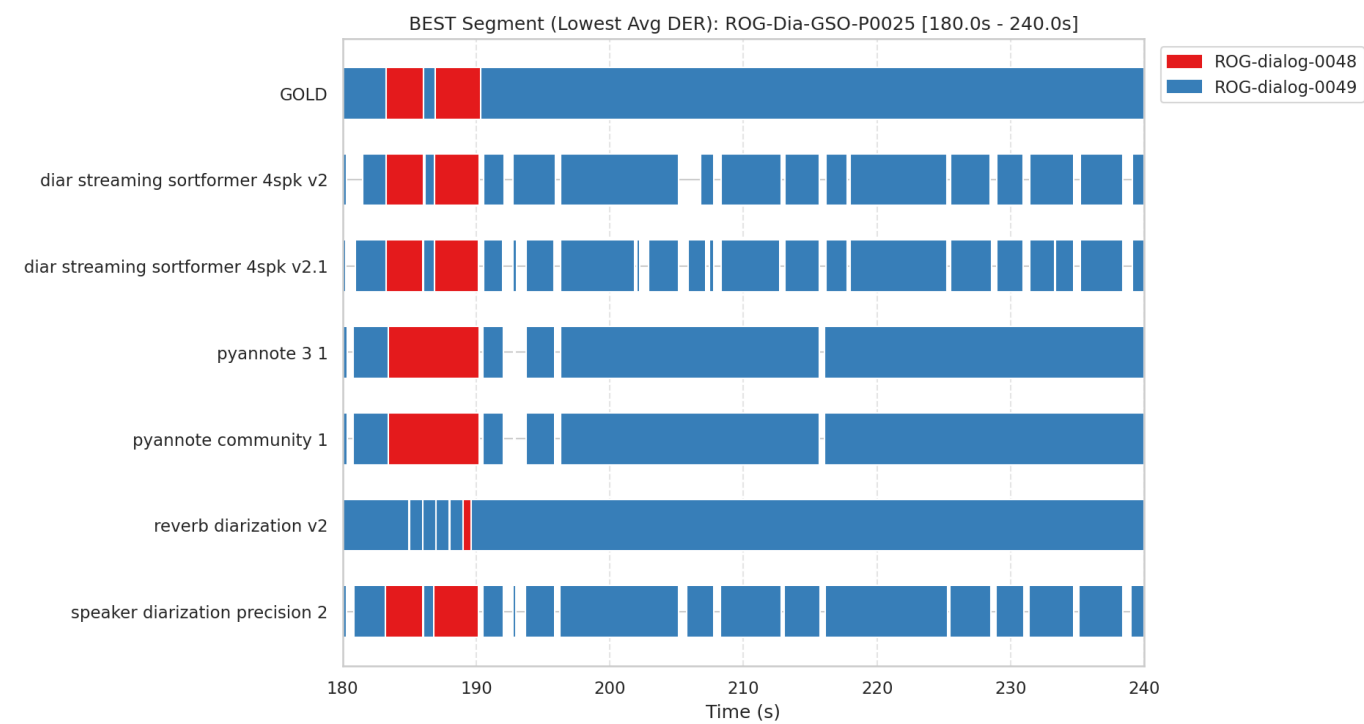
Domain: Navodila | **Quality:** 5 | **Device:** prenosni računalnik acer predator helios 300 z zunanjim mikrofonom RODE NT-USB

Pogovor o potovanju na Japonsko, iskanju novega stanovanja, službi in pripravi raznih Japonskih jedi



60-Second Snippets (Zoom-in)

Below are 60-second zoomed-in windows showing where the models performed best and worst (based on average DER).





Model	DER	Miss	FA	Conf	Pur	Cov	VRAM (GB)	Status
reverb diarization v2	13.85	3.45	5.99	4.42	91.66	91.60	4.3	nan
pyannote community 1	15.84	14.24	0.82	0.78	94.60	95.68	1.6	nan
pyannote 3 1	15.88	14.24	0.82	0.82	94.55	95.41	1.6	nan
speaker diarization precision 2	19.93	18.88	0.77	0.28	94.77	95.93	0.0	nan
diar streaming sortformer 4spk v2	22.58	21.16	1.23	0.19	94.38	95.67	1.6	nan
diar streaming sortformer 4spk v2.1	26.30	23.21	0.92	2.17	94.85	93.87	1.6	nan
diar sortformer 4spk v1	nan	nan	nan	nan	nan	nan	nan	OOM/ERR